

The limits of fair medical imaging AI in real-world generalization

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Yuzhe Yang^{1,4}✉, Haoran Zhang^{1,4}, Judy W. Gichoya², Dina Katabi¹ & Marzyeh Ghassemi^{1,3}

As artificial intelligence (AI) rapidly approaches human-level performance in medical imaging, it is crucial that it does not exacerbate or propagate healthcare disparities. Previous research established AI's capacity to infer demographic data from chest X-rays, leading to a key concern: do models using demographic shortcuts have unfair predictions across subpopulations? In this study, we conducted a thorough investigation into the extent to which medical AI uses demographic encodings, focusing on potential fairness discrepancies within both in-distribution training sets and external test sets. Our analysis covers three key medical imaging disciplines—radiology, dermatology and ophthalmology—and incorporates data from six global chest X-ray datasets. We confirm that medical imaging AI leverages demographic shortcuts in disease classification. Although correcting shortcuts algorithmically effectively addresses fairness gaps to create 'locally optimal' models within the original data distribution, this optimality is not true in new test settings. Surprisingly, we found that models with less encoding of demographic attributes are often most 'globally optimal', exhibiting better fairness during model evaluation in new test environments. Our work establishes best practices for medical imaging models that maintain their performance and fairness in deployments beyond their initial training contexts, underscoring critical considerations for AI clinical deployments across populations and sites.

As artificial intelligence (AI) models are increasingly deployed in real-world clinical settings^{1,2}, it is crucial to evaluate not only model performance but also potential biases toward specific demographic groups^{3,4}. Although deep learning has achieved human-level performance in numerous medical imaging tasks^{5,6}, existing literature indicates a tendency for these models to manifest existing biases in the data, causing performance disparities between protected subgroups^{7–11}. For instance, chest X-ray (CXR) classifiers trained to predict the presence of disease systematically underdiagnose Black patients¹², potentially leading to delays in care. To ensure the responsible and equitable deployment of such models, it is essential to understand the source of such biases and, where feasible, take actions to correct them^{13,14}.

Recent studies have unveiled the surprising ability of deep models to predict demographic information, such as self-reported race¹⁵, sex and age¹⁶, from medical images, achieving performance far beyond that of radiologists. These insights raise the concern of disease prediction models leveraging demographic features as heuristic 'shortcuts'^{17,18}—correlations that are present in the data but have no real clinical basis¹⁸, for instance deep models using the hospital as a shortcut for disease prediction^{19,20}.

In this work, we investigated four questions. First, we consider whether disease classification models also use demographic information as shortcuts and whether such demographic shortcuts result in biased predictions. Second, we evaluate the extent to which

¹Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, Cambridge, MA, USA. ²Department of Radiology, Emory University School of Medicine, Atlanta, GA, USA. ³Institute for Medical Engineering & Science, Massachusetts Institute of Technology, Cambridge, MA, USA. ⁴These authors contributed equally: Yuzhe Yang, Haoran Zhang. ✉e-mail: yuzhe@mit.edu

state-of-the-art methods can remove such shortcuts and create ‘locally optimal’ models that are also fair. Third, we consider real-world clinical deployment settings where shortcuts may not be valid in the out-of-distribution (OOD) data, to dissect the interplay between algorithmic fairness and shortcuts when data shift. Finally, we explore which algorithms and model selection criteria can lead to ‘globally optimal’ models that maintain fairness when deployed in an OOD setting.

We performed a systematic investigation into how medical AI leverages demographic shortcuts through these questions, with an emphasis on fairness disparities across both in-distribution (ID) training and external test sets. Our primary focus is on CXR prediction models, with further validation in dermatology (Extended Data Fig. 1) and ophthalmology (Extended Data Fig. 2). Our X-ray analysis draws upon six extensive, international radiology datasets: MIMIC-CXR²¹, CheXpert²², NIH²³, SIIM²⁴, PadChest²⁵ and VinDr²⁶. We explored fairness within both individual and intersectional subgroups spanning race, sex and age¹². Our assessment uncovers compelling new insights into how medical AI encodes demographics and the impact that this has on various fairness considerations, especially when models are applied outside their training context during real-world domain shifts, with actionable insights on what models to select for fairness under distribution shift.

Results

Datasets and model training

We used six publicly available CXR datasets, as described in Table 1. We focused on four binary classification tasks that have been shown to have disparate performance between protected groups^{7,27}: ‘No Finding’, ‘Effusion’, ‘Pneumothorax’ and ‘Cardiomegaly’. The detailed prevalence rates of the diseases for each demographic subgroup are shown in Extended Data Table 1.

We also examined medical AI applications in dermatology and ophthalmology. Specifically, we used the ISIC dataset²⁸ with ‘No Finding’ as the task for dermatological imaging (Extended Data Fig. 1a) and the ODIR dataset²⁹ with ‘Retinopathy’ as the task for ophthalmology images (Extended Data Fig. 2a).

To evaluate fairness, we examined the class-conditioned error rate that is likely to lead to worse patient outcomes for a screening model. For ‘No Finding’, a false positive indicates falsely predicting that a patient is healthy when they are ill, which could lead to delays in treatment¹²; we, therefore, evaluated the differences in false-positive rate (FPR) between demographic groups. For all other diseases, we evaluated the false-negative rate (FNR) for the same reason. Equality in these metrics is equivalent to equality of opportunity³⁰. We choose to study fairness through the notion of equalized odds, as it has been widely used in previous work in the CXR and fairness literature^{7,12}. In addition, shortcut learning using a particular demographic attribute leads to differences in class-conditioned error rates (that is, FPR and FNR gaps) across attributes^{31,32}, and so studying these gaps allows us to glean insight into the severity of shortcut learning. Finally, FPR and FNR (as enforced to be equal by equalized odds) are meaningful metrics in the clinical setting, as they correspond to error rates of decision-making at the individual level¹².

To understand and quantify the types and degrees of distribution shifts in our study, we examined whether there are significant statistical differences in distributions between demographic groups in the ID settings as well as across different datasets in the OOD settings. Specifically, we analyzed prevalence shifts $P(Y|A)$ and representation shifts $P(X|A)$ across different subgroups for ID scenarios and added label shifts $P(Y)$ and covariate shifts $P(X)$ for OOD scenarios (Methods). Our analyses indicate that all the distributions that we examined show statistically significant shifts, affecting most demographic groups in the ID context (Extended Data Table 2) and across various sites in the OOD context (Extended Data Table 3). We note that our analysis does not presuppose specific types of distribution shifts; instead, we

simulated real-world deployment conditions where any of these shifts might occur, aiming for results that are generalizable to complex, real-world scenarios.

We trained a grid of deep convolutional neural networks³³ on MIMIC-CXR (radiology), CheXpert (radiology), ODIR (ophthalmology) and ISIC (dermatology), varying the classification task. Our approach follows previous work that achieves state-of-the-art performance in these tasks^{8,12} using empirical risk minimization (ERM)³⁴. We also evaluated algorithms designed to remove spurious correlations or increase model fairness during training. We categorized these algorithms into those that (1) reweight samples based on their group to combat underrepresentation (ReSample³⁵ and GroupDRO³⁶); (2) adversarially remove group information from model representations (DANN³⁷ and CDANN³⁸); and (3) more generically attempt to improve model generalization—that is, exponential moving average (MA)³⁹. In total, our analysis encompassed a total of 3,456 models trained on MIMIC-CXR, corresponding to the cartesian product of four tasks, four demographic attributes, six algorithms, 12 hyperparameter settings and three random seeds. We summarized our experimental pipeline in Fig. 1.

Algorithmic encoding of attributes leads to fairness gaps

We separately trained deep learning models for our four distinct CXR prediction tasks (‘No Finding’, ‘Cardiomegaly’, ‘Effusion’ and ‘Pneumothorax’) as well as ‘Retinopathy’ in ophthalmology and ‘No Finding’ in dermatology. Each model consists of a feature extractor followed by a disease prediction head. We then employed a transfer learning approach, wherein we kept the weights of the feature extractor frozen and retrained the model to predict sensitive attributes (for example, race). This allowed us to assess the amount of attribute-related information present in the features learned by each model as measured by the area under the receiver operating characteristic curve (AUROC) for attribute prediction (Methods). Previous work^{15,40} demonstrated that deep models trained for disease classification encode demographic attributes, and such encoding could lead to algorithmic bias⁴¹. We extend the investigation to a broader array of datasets, attributes and imaging modalities. As Fig. 2a,c,e confirms, the penultimate layer of different disease models contains significant information about four demographic attributes (age, race, sex and the intersection of sex and race), and that is consistent across different tasks and medical imaging modalities.

We then assessed the fairness of these models across demographic subgroups as defined by equal opportunity³⁰—that is, discrepancies in the model’s FNR or FPR for demographic attributes. We focused on underdiagnosis¹²—that is, discrepancies in FPR for ‘No Finding’ and discrepancies in FNR for other diseases. For each demographic attribute, we identified two key subgroups with sufficient sample sizes: age groups ‘80–100’ ($n = 8,063$) and ‘18–40’ ($n = 7,319$); race groups ‘White’ ($n = 32,732$) and ‘Black’ ($n = 8,279$); sex groups ‘female’ ($n = 25,782$) and ‘male’ ($n = 27,794$); and sex and race groups ‘White male’ ($n = 18,032$) and ‘Black female’ ($n = 5,027$). In all tasks, we observed that the models displayed biased performance within the four demographic attributes, as evidenced by the FNR disparities (Fig. 2b). The observed gaps can be as large as 30% for age. The same results hold for the other two imaging modalities (Fig. 2d,f). Similar results for overdiagnosis (FNR of ‘No Finding’ and FPR for disease prediction) can be found in Extended Data Fig. 3.

We further investigated the degree to which demographic attribute encoding ‘shortcuts’ may impact model fairness. When models use demographic variables as shortcuts, previous work showed that they can exhibit gaps in subgroup FPR and FNR^{31,40}. We note that a model encoding demographic information does not necessarily imply a fairness violation, as the model may not necessarily use this information for its prediction. For each task and attribute combination, we trained different models with varying hyperparameters (Methods). We focused on the correlation between the degree of encoding of

Table 1 | Demographic and label characteristics of the six X-ray datasets used in this study

	Location	MIMIC	CheXpert	NIH	SIIM	PadChest	VinDr
		Boston, MA	Stanford, CA	Bethesda, MD	Bethesda, MD	Alicante, Spain	Hanoi, Vietnam
		No. of images	222,792	112,120	11,582	144,478	6,354
		Percent frontal	85.5	100.0	100.0	69.1	100.0
Sex (%)	Female	47.8	40.7	43.5	44.6	50.4	43.1
	Male	52.2	59.3	56.5	55.4	49.6	56.9
Race (%)	Asian	3.1	10.5	–	–	–	–
	Black	15.6	5.4	–	–	–	–
	White	61.0	56.4	–	–	–	–
	Other	20.3	27.8	–	–	–	–
Age (%)	0–18	–	–	4.8	5.0	3.7	21.8
	18–40	13.8	13.9	27.7	27.3	9.2	16.0
	40–60	31.1	31.1	43.9	42.9	26.5	27.1
	60–80	40.0	39.0	22.7	23.9	38.0	30.0
	80–100	15.1	16.0	0.9	0.9	22.6	5.1
Intersection (%)	Asian female	1.5	4.5	–	–	–	–
	Asian male	1.6	6.0	–	–	–	–
	Black female	9.3	2.6	–	–	–	–
	Black male	6.3	2.7	–	–	–	–
	White female	27.3	22.2	–	–	–	–
	White male	33.8	34.1	–	–	–	–
	Others female	9.8	11.3	–	–	–	–
	Others male	10.5	16.5	–	–	–	–
Task prevalence (%)	No Finding	39.8	10.0	53.8	–	34.9	41.2
	Effusion	20.0	38.6	11.9	–	5.9	7.5
	Pneumothorax	3.4	8.7	4.7	28.4	0.3	0.7
	Cardiomegaly	14.9	12.1	2.5	–	9.5	22.6

different attributes and the fairness gaps as assessed by underdiagnosis. Figure 2g shows that a stronger encoding of demographic information is significantly correlated with stronger model unfairness ('No Finding', age: $R = 0.82$, $P = 4.7 \times 10^{-8}$; 'No Finding', sex and race: $R = 0.81$, $P = 8.4 \times 10^{-9}$; 'Cardiomegaly', age: $R = 0.81$, $P = 1.9 \times 10^{-7}$; 'Effusion', race: $R = 0.71$, $P = 6.4 \times 10^{-6}$; 'Pneumothorax', sex: $R = 0.59$, $P = 2.3 \times 10^{-3}$; all using two-sided t -test). Such consistent observations indicate that models using demographic encodings as heuristic shortcuts also have larger fairness disparities, as measured by discrepancies in FPR and FNR.

Mitigating shortcuts creates locally optimal models

We performed model evaluations first in the ID setting, where ERM models trained and tested on data from the same source performed well. We compared ERM to state-of-the-art robustness methods that were designed to effectively address fairness gaps while maintaining overall performance. As shown in Fig. 3a, ERM models exhibited large fairness gaps across age groups when predicting 'Cardiomegaly' (that is, models centered in the top right corner, FNR gap of 20% between groups '80–100' and '18–40'). By applying data rebalancing methods to address prevalence shifts during training (for example, ReSample), we observed reduced fairness gaps in certain contexts. By applying debiasing robustness methods that correct demographic shortcuts, such as GroupDRO and DANN, the resulting models were able to close the FNR gap while achieving similar AUROCs (for example, the bottom right corner). Our results hold when using the worst group AUROC as the performance metric (Extended Data Fig. 4) and across different combinations of diseases and attributes (Fig. 3b and Extended Data Fig. 4).

To demonstrate the value of model debiasing, we further plotted the set of 'locally optimal models'—those on the Pareto front⁴² that balance the performance–fairness tradeoff most optimally on ID data (Fig. 3a). Those models that lie on this front are 'locally optimal', as they have the smallest fairness gap that can be achieved for a fixed performance constraint (for example, AUROC > 0.8). In the ID setting, we found several existing algorithms that consistently achieve high ID fairness without losing notable overall performance for disease prediction (Fig. 3a,b and Extended Data Fig. 4).

Similar to our observations in radiology, we identified fairness gaps within subgroups based on age and sex in dermatology and ophthalmology, respectively (Fig. 2d,f). We further verified the Pareto front for both attributes, where similar observations hold that algorithms for fixing demographic shortcuts could improve ID fairness while incurring minimal detriments to performance (Fig. 3c,d). The steepness of the Pareto front suggests that small sacrifices in performance could yield substantial gains in fairness.

Locally optimal models exhibit tradeoffs in other metrics

We examined how locally optimal models that balance fairness and AUROC impact other metrics, as previous work showed that it is a theoretical impossibility to balance fairness measured by probabilistic equalized odds and calibration by group⁴³. We found that optimizing fairness alone leads to worse results for other clinically meaningful metrics in some cases, indicating an inherent tradeoff between fairness and other metrics. First, for the 'No Finding' prediction task, enforcing fair predictions across groups results in worse expected calibration error gap (ECE Gap; Extended Data Fig. 5a) between groups. Across different

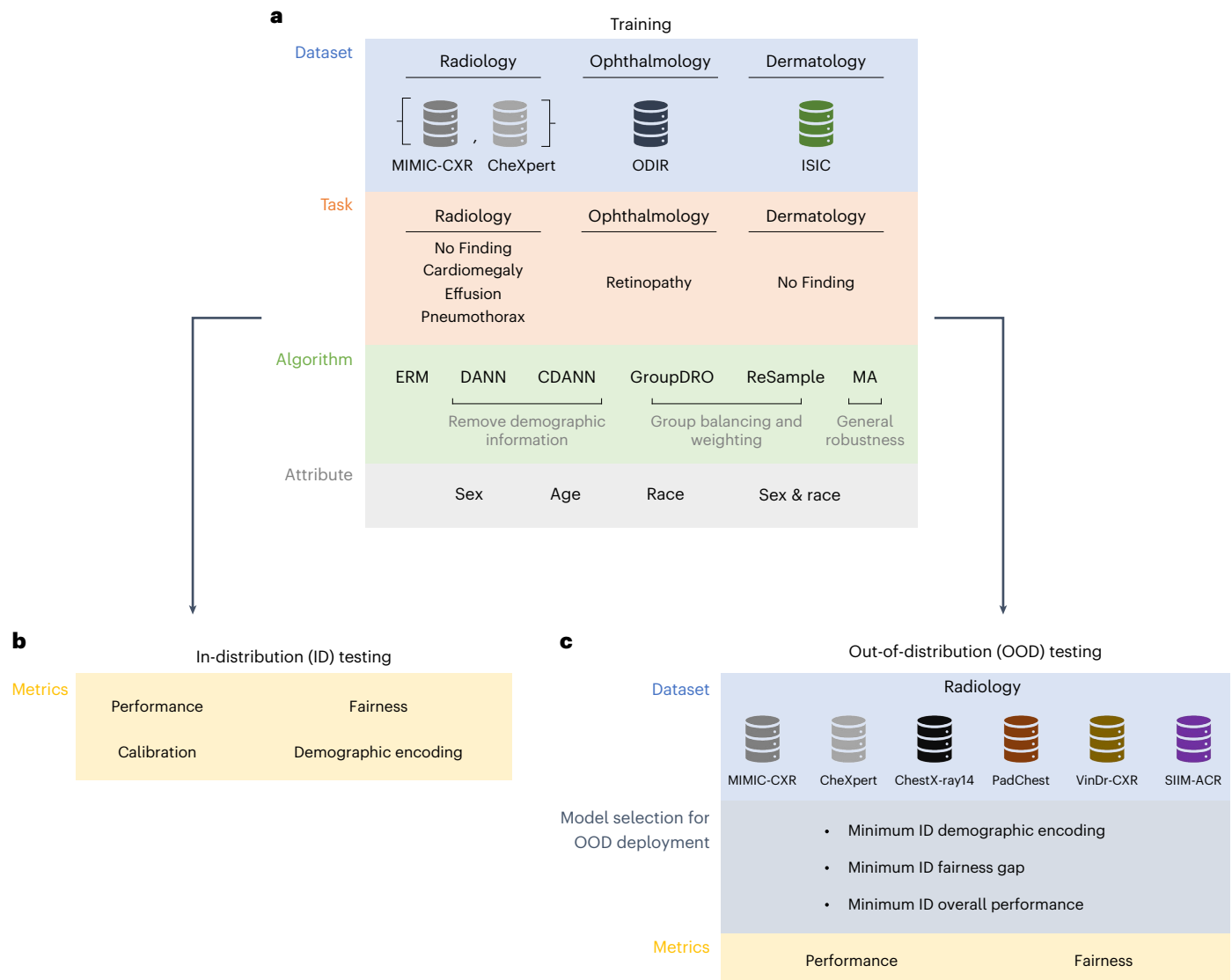


Fig. 1 | Overall experimental pipeline. a, We trained a grid of deep learning models on medical images from a variety of modalities on several clinical tasks. We applied a variety of state-of-the-art algorithms to mitigate shortcuts, for up to four demographic attributes (where available). **b**, We evaluated each model ID (that is, on the same dataset where it is trained), along the axis of performance,

fairness, amount of demographic encoded and calibration. **c**, We evaluated the performance and fairness of CXR classification models on OOD domains. To mimic a realistic deployment setting where OOD samples are not observed, we chose the ‘best’ model based on several ID selection criteria.

demographic attributes, we found a consistent statistically significant negative correlation between ECE Gap and Fairness Gap (age: $R = -0.85$, $P = 7.5 \times 10^{-42}$; race: $R = -0.64$, $P = 6.1 \times 10^{-15}$; sex: $R = -0.73$, $P = 4.4 \times 10^{-28}$; sex and race: $R = -0.45$, $P = 1.9 \times 10^{-8}$; all using two-sided t -test).

We explored the relationship between fairness and other metrics, including average precision and average F1 score. For ‘No Finding’ prediction, fairer models lead to both worse average precision and F1 score (Extended Data Fig. 5a). The same trend holds across different diseases—for example, for ‘Effusion’ (Extended Data Fig. 5b). These findings stress that these models, although being locally optimal, exhibit worse results on other important and clinically relevant performance metrics. This uncovers the limitation of blindly optimizing fairness, emphasizing the necessity for more comprehensive evaluations to ensure the reliability of medical AI models.

Local fairness does not transfer under distribution shift

When deploying AI models in real settings, it is crucial to ensure that models can generalize to data from unseen institutions or environments. We directly tested all trained models in the OOD setting, where

we report results on external test datasets that are unseen during model development. Figure 4 illustrates that the correlation between ID and OOD performance is high across different settings, which was observed in previous work⁴⁴. However, we found that there was no consistent correlation between ID and OOD fairness. For example, Fig. 4b shows an instance where the correlation between ID fairness and OOD fairness is strongly positive (‘Effusion’ with ‘age’ as the attribute; $R = 0.98$, $P = 3.0 \times 10^{-36}$, two-sided t -test), whereas Fig. 4c shows an instance where the correlation between these metrics is actually significantly negative (‘Pneumothorax’ with ‘sex and race’ as the attribute; $R = -0.50$, $P = 4.4 \times 10^{-3}$, two-sided t -test). Across 16 combinations of task and attribute, we found that five such settings exhibited this negative correlation, and three additional settings exhibited only a weak ($R < 0.5$) positive correlation (see Extended Data Fig. 6a,b for additional correlation plots). Thus, improving ID fairness may not lead to improvements in OOD fairness, highlighting the complex interplay between fairness and distribution shift^{45,46}.

In addition, we investigated whether models achieving ID Pareto optimality between fairness and performance will maintain in OOD

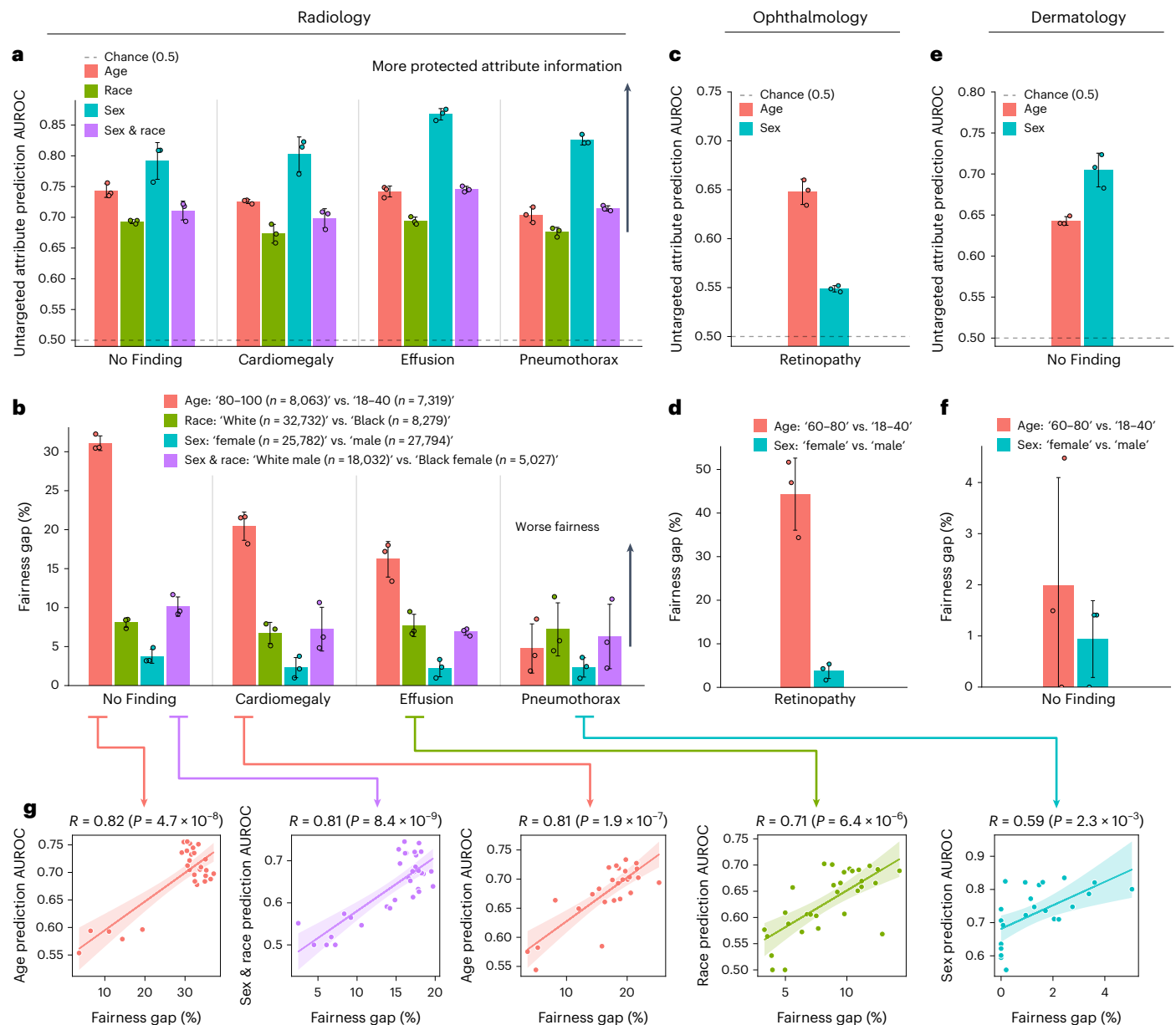


Fig. 2 | Medical imaging models encode sensitive attributes and are unfair across subgroups. **a**, The AUROC of demographic attribute prediction from frozen representations for the best ERM model. We trained ERM models on MIMIC-CXR to predict four different binary tasks. ERM representations encode demographic attributes to a high degree. **b**, The fairness gap, as defined by the FPR gap for 'No Finding', and the FNR gap for all other tasks for the best ERM model. ERM models exhibit high fairness gaps, especially between age groups. **c**, The AUROC of demographic attribute prediction from frozen representations for the best ERM model on the ODIR dataset (ophthalmology), following the same experimental setup. **d**, The fairness gap for the best ERM model on the ODIR dataset (ophthalmology). **e**, The AUROC of demographic attribute prediction from frozen representations for the best ERM model on the ISIC dataset

(dermatology), following the same experimental setup. **f**, The fairness gap for the best ERM model on the ISIC dataset (dermatology). **a-f**, Each bar and its error bar indicate the mean and standard deviation across three independent runs. **g**, The correlation between attribute prediction performance and fairness for all learned models. We excluded models with suboptimal performance—that is, with an overall validation AUROC below 0.7. The attribute prediction AUROC shows a high correlation with the fairness gap ('No Finding', age: $R = 0.82$, $P = 4.7 \times 10^{-8}$; 'No Finding', sex and race: $R = 0.81$, $P = 8.4 \times 10^{-9}$; 'Cardiomegaly', age: $R = 0.81$, $P = 1.9 \times 10^{-7}$; 'Effusion', race: $R = 0.71$, $P = 6.4 \times 10^{-6}$; 'Pneumothorax', sex: $R = 0.59$, $P = 2.3 \times 10^{-3}$; all using two-sided t -test). The center line and the shadow denote the mean and 95% CI, respectively.

settings. As shown for 'Cardiomegaly' prediction using race as the attribute, models originally on the Pareto front ID (Fig. 4d) do not guarantee to maintain Pareto optimality when deployed in a different OOD setting (Fig. 4e). We show additional examples of this phenomenon in Extended Data Fig. 6c.

Dissecting model fairness under distribution shift

To disentangle the OOD fairness gap, we present a way to decompose model fairness under distribution shift. Specifically, we decompose

and attribute the change in fairness between ID and OOD to be the difference in performance change for each of the groups—that is, the change in fairness is determined by how differently the distribution shift affects each group (Methods).

In Extended Data Fig. 7, we show examples of transferring a trained model from ID setting to OOD setting. For example, Extended Data Fig. 7d illustrates an ERM model trained to predict 'No Finding' on CheXpert (ID) and transferred to MIMIC-CXR (OOD) while evaluating fairness across sex. We found that the model was fair with respect to

Radiology

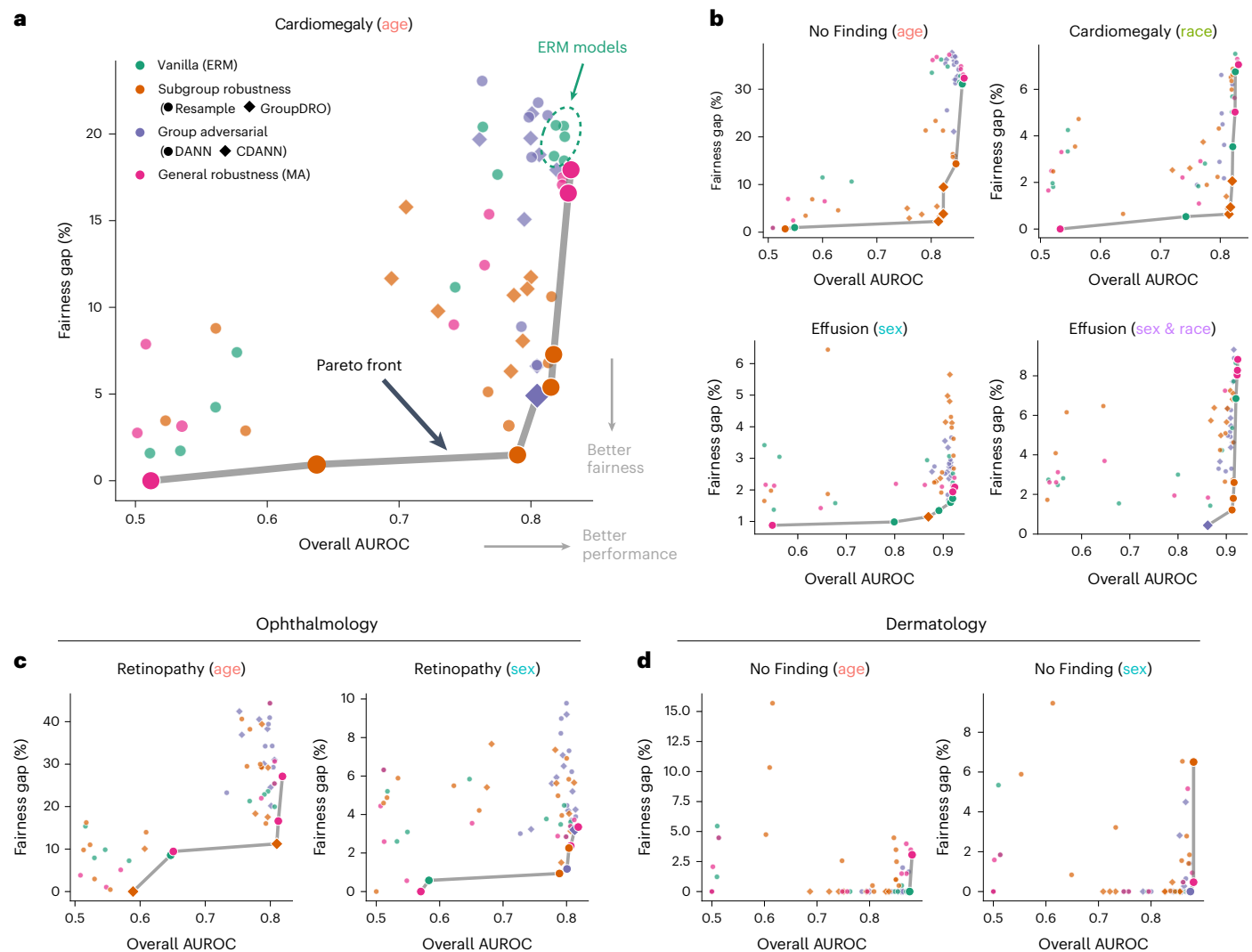


Fig. 3 | Algorithms for removing demographic shortcuts mitigate ID fairness gaps and maintain performance. a, Tradeoff between the fairness gap and overall AUROC for all trained models, for ‘Cardiomegaly’ prediction using ‘age’ as the attribute. We plotted the Pareto front—the best achievable fairness gap with a minimum constraint on the performance. **b**, Tradeoff between the fairness gap

and overall AUROC for all trained models, with more disease prediction tasks and attributes. **c**, Tradeoff between the fairness gap and the overall AUROC on the ODIR dataset (ophthalmology). **d**, Tradeoff between the fairness gap and the overall AUROC on the ISIC dataset (dermatology).

the FPR gap in the ID setting (-0.1% gap, not significant) but had a significant FPR gap when deployed in the OOD setting (3.2%), with females being underdiagnosed at a higher rate (Extended Data Fig. 7e). We then segmented this FPR gap by sex and found that females experienced an increase in FPR of 3.9% , whereas males experienced an increase in FPR of 0.8% (Extended Data Fig. 7f). In other words, the model becomes worse for both groups in an OOD setting but to a much larger extent for female patients. This decomposition suggests that mitigation strategies that reduce the impact of the distribution shift on females could be effective in reducing the OOD fairness gap in this instance.

We further extended this study to a larger set of tasks and protected attributes (Extended Data Fig. 7). Across all settings, the disparate impact of distribution shift on each group was a significant component, indicating that mitigating the impact of distribution shift is as important as mitigating ID fairness, if the goal is to achieve a fair model OOD.

Globally optimal model selection for OOD fairness

Figure 4 shows that selecting a model based on ID fairness may not lead to a model with optimal OOD fairness. Here, we examined alternate

model selection criteria that may lead to better OOD fairness, when we have access only to ID data. Our goal is to find ‘globally optimal’ models that maintain their performance and fairness in new domains. First, we subsetting our selection only to models that had satisfactory ID overall performance (defined as those with overall validation AUROC no less than 5% of the best ERM model). This set of models also had satisfactory OOD performance (Supplementary Fig. 1).

Next, we proposed eight candidate model selection criteria (Fig. 5a), corresponding to selecting the model from this set that minimizes or maximizes some ID metric. We evaluated the selected model by its OOD fairness across five external datasets, each containing up to four attributes and up to four tasks, corresponding to a total of 42 settings. We compared the OOD fairness of the selected model to the OOD fairness of an ‘oracle’, which observes samples from the OOD dataset and directly chooses the model with the smallest OOD fairness gap. For each setting, we computed the increase in fairness gap of each selection criteria relative to the oracle. In Fig. 5a, we report the mean across the 42 settings as well as the 95% confidence interval (CI) computed from 1,000 bootstrap iterations. We found that, surprisingly, selecting the

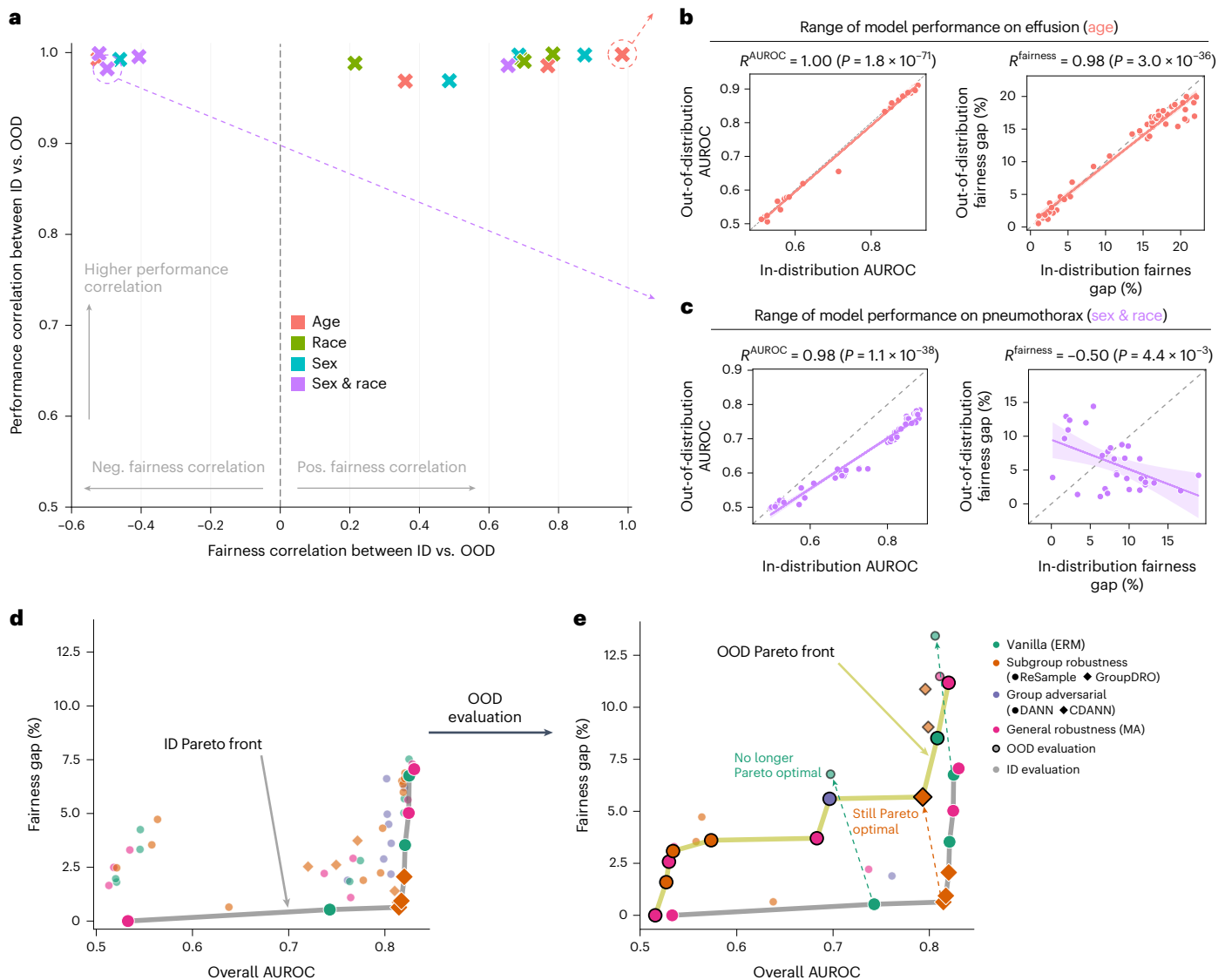


Fig. 4 | The transfer of performance (overall AUROC) and fairness between the ID (MIMIC-CXR) and OOD datasets. **a**, We plotted the Pearson correlation coefficient of ID versus OOD performance versus the Pearson correlation coefficient of ID versus OOD fairness. Here, each point was derived from a grid of models trained on a particular combination of task and attribute. We found that there was a high correlation between ID and OOD performance in all cases, but the correlation between ID and OOD fairness was tenuous. **b**, One particular point where fairness transfers between ID and OOD datasets ('Effusion' with 'age' as the attribute; $R = 0.98$, $P = 3.0 \times 10^{-36}$, two-sided t -test). The center line and

the shadow denote the mean and 95% CI, respectively. **c**, One particular point where fairness does not transfer between ID and OOD datasets ('Pneumothorax' with 'sex and race' as the attribute; $R = -0.50$, $P = 4.4 \times 10^{-3}$, two-sided t -test). The center line and the shadow denote the mean and 95% CI, respectively. **d**, The ID Pareto front for 'Cardiomegaly' prediction using 'race' as the attribute. **e**, The transformation of the ID Pareto front to the OOD Pareto front, for 'Cardiomegaly' prediction using 'race' as the attribute. Models that are Pareto optimal ID often do not maintain Pareto optimality OOD.

model with the minimum ID fairness gap may not be optimal. Instead, two other criteria based on selecting models where the embedding contains the least attribute information lead to a lower average OOD fairness gap. For instance, we observed a significantly lower increase in OOD fairness gap by selecting models with the 'Minimum Attribute Prediction Accuracy' as compared to 'Minimum Fairness Gap' ($P = 9.60 \times 10^{-94}$, one-tailed Wilcoxon rank-sum test). The result echoes our finding in Fig. 2 that the encoding of demographic attributes is positively correlated with ID fairness.

Finally, we studied the fairness of each algorithm in the OOD setting. We maintained the performance cutoff described above and selected the model for each algorithm with the lowest ID fairness gap. In Fig. 5b, we report the mean increase in OOD fairness gap relative to the oracle across the same 42 settings. We found that methods that remove demographic information from embeddings (specifically,

DANN) lead to the lowest average OOD fairness gap ('DANN' versus 'ERM': $P = 1.86 \times 10^{-117}$, one-tailed Wilcoxon rank-sum test). Our findings demonstrate that evaluating and removing demographic information encoded by the model ID may be the key to 'globally optimal' models that transfer both performance and fairness to external domains.

Discussion

We demonstrated the interplays between the demographic encoding of attributes as 'shortcuts' in medical imaging AI models and how they change under distribution shifts. Notably, we validated our findings across global-scale datasets in radiology (Table 1) and across multiple medical imaging modalities (Extended Data Figs. 1 and 2). The results show that algorithmic encoding of protected attributes leads to unfairness (Fig. 2) and mitigating shortcuts can reduce ID fairness gaps and maintain performance (Fig. 3). However, our results also show that there

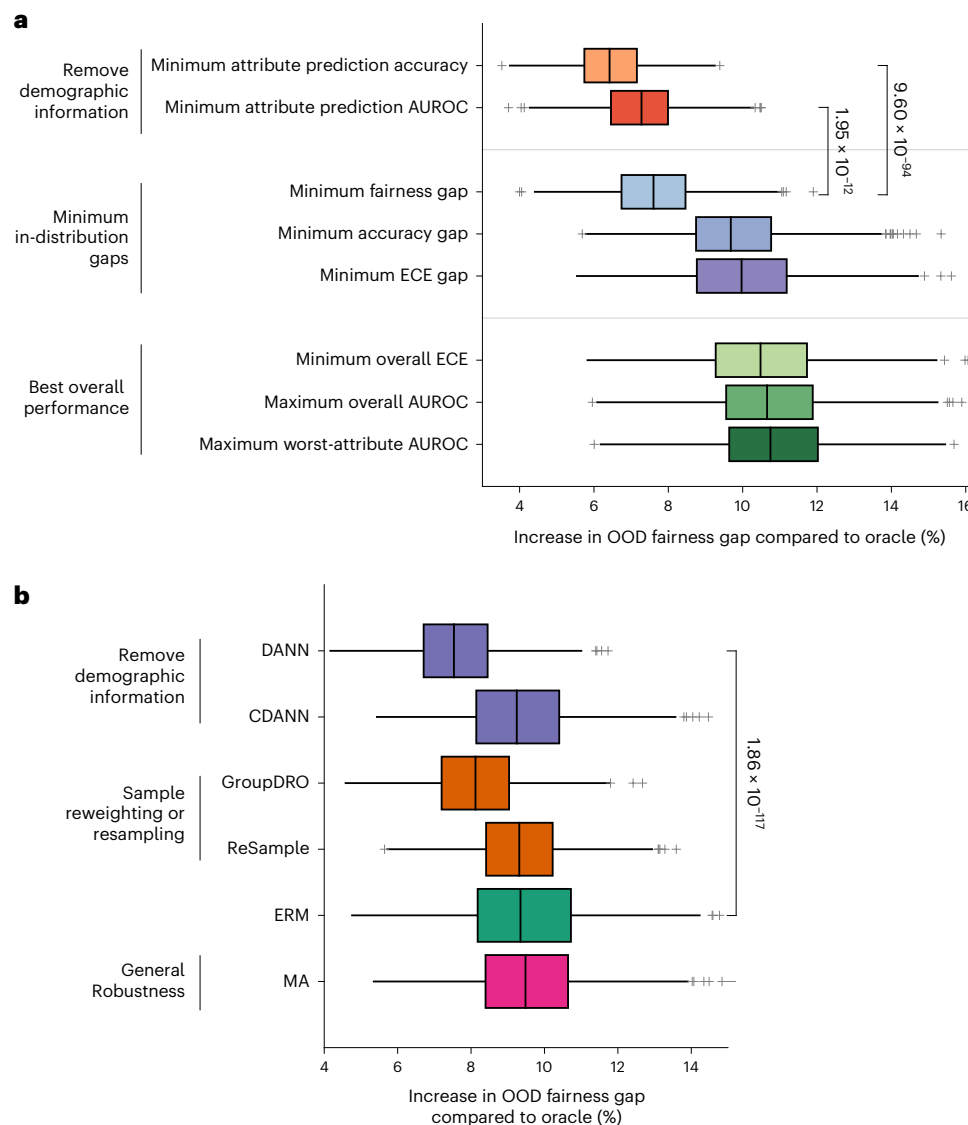


Fig. 5 | OOD fairness of models with different model selection criteria and for different algorithms. a, We varied the ID model selection criteria and compared the selected model against the oracle that chooses the model that is most fair OOD. We plotted the increase in OOD fairness gap of the selected model over the oracle, averaged across 42 combinations of OOD dataset, task and attribute. We used non-parametric bootstrap sampling ($n = 1,000$) to define the bootstrap distribution for the metric. We found that selection criteria based on choosing models with minimum attribute encoding achieve better OOD fairness than naively selecting based on ID fairness or other aggregate performance metrics ('Minimum Attribute Prediction Accuracy' versus 'Minimum Fairness

Gap': $P = 9.60 \times 10^{-94}$, one-tailed Wilcoxon rank-sum test; 'Minimum Attribute Prediction AUROC' versus 'Minimum Fairness Gap': $P = 1.95 \times 10^{-12}$, one-tailed Wilcoxon rank-sum test). **b**, We selected the model for each algorithm with the minimum ID fairness gap. We evaluated its OOD fairness against the oracle on the same 42 settings. We found that removing demographic encoding (that is, DANN) leads to the best OOD fairness ('DANN' versus 'ERM': $P = 1.86 \times 10^{-17}$, one-tailed Wilcoxon rank-sum test). On each box, the central line indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The whiskers extend to 1.5 times the interquartile range. Points beyond the whiskers are plotted individually using the '+' symbol.

exists an inherent tradeoff for clinically meaningful metrics beyond fairness (Extended Data Fig. 5), and such fairness does not transfer under distribution shift (Fig. 4). We provide initial strategies to dissect and explain the model fairness under distribution shifts (Extended Data Fig. 7). Our results further reveal actionable algorithm and model selection strategies for OOD fairness (Fig. 5).

Our results have multiple implications. First, they offer a cautionary tale on the efficacy and consequences of eliminating demographic shortcuts in disease classification models. On the one hand, removing shortcuts addresses ID fairness, which is a crucial consideration in fair clinical decision-making¹². On the other hand, the resulting tradeoffs with other metrics and non-transferability to OOD settings raises the question about the long-term utility in removing such shortcuts. This is particularly complex in the healthcare setting, where the relationship

between demographics and the disease or outcome label is complex⁴⁷, variables can be mislabeled⁴⁸ and distribution shifts between domains are difficult to quantify¹.

Second, we frame demographic features as potential 'shortcuts', which should not be used by the model to make disease predictions. However, some demographic variables could be a direct causal factor in some diseases (for example, sex as a causal factor of breast cancer). In these cases, it would not be desirable to remove all demographic reliance but, instead, match the reliance of the model on the demographic attribute to its true causal effect⁴⁹. In the tasks that we examined here, demographic variables, such as race, may have an indirect effect on disease (for example, through socioeconomic status)⁵⁰, which may vary across geographic location or even time period⁵¹. Whether demographic variables should serve as proxies

for these causal factors is a decision that should rest with the model deployers^{14,47,52,53}.

Third, we present a preliminary decomposition for diagnosing OOD model fairness changes, by expressing it as a function of the ID fairness gap and the performance change of each group. We found that the disparate impact of distribution shift on per-group performance is a major contributor to lack of fairness in OOD settings. Our work suggests that, for practitioners trying to achieve fairness in models deployed in a different domain, mitigating ID fairness is at least as important as mitigating the impact of distribution shift for particular groups. However, building models robust to arbitrary domain shifts is, in general, a challenging task^{54,55}. Having some knowledge or data about how the distributions may shift, or even the ability to actively collect data for particular groups, may be necessary⁵⁶. Developing methods and deriving theoretical characterizations of fairness under distribution shift is an active area of research^{45,46}.

Fourth, the US Food & Drug Administration (FDA), as the primary regulatory body for medical technologies, does not require external validation of clinical AI models, relying instead on the assessment by the product creator⁵⁷. Our findings underscore the necessity for regular evaluation of model performance under distribution shift^{58,59}, challenging the popular opinion of a single fair model across different settings⁶⁰. This questions the effectiveness of developer assurances on model fairness at the time of testing and highlights the need for regulatory bodies to consider real-world performance monitoring, including fairness degradation⁶¹. Finally, when a model is deployed in any clinical environment, both its overall and per-group performance, as well as associated clinical outcomes, should be continuously monitored⁶².

Finally, although we imply that smaller ‘fairness gaps’ are better, enforcing these group fairness definitions can lead to worse utility and performance for all groups^{43,63–66}, and other fairness definitions may be better suited to the clinical setting^{8,67}. We note that these invariant notions of fairness could have drawbacks⁶⁶, as equalized odds are incompatible with calibration by group (Extended Data Fig. 5), and enforcing equalized odds often lead to the ‘leveling down’ effect in overall performance^{63,64}. We present the Pareto curve showing the tradeoff between fairness and accuracy, allowing the practitioner to select a model that best fits their deployment scenario. In general, we encourage practitioners to choose a fairness definition that is best suited to their use case and carefully consider the performance–equality tradeoff. The impact of minimizing algorithmic bias on real-world health disparities, the ultimate objective, is complex⁶⁸, and there is no guarantee that deploying a fair model will lead to equitable outcomes. In addition, although we constructed several models for clinical risk prediction, we do not advocate for deployment of these models in real-world clinical settings without practitioners carefully testing models on their data and taking other considerations into account (for example, privacy, regulation and interpretability)^{1,3}.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-024-03113-4>.

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Methods

Datasets and pre-processing

The datasets used in this study are summarized in Extended Data Table 1. Unless otherwise stated, we trained models on MIMIC-CXR²¹ and evaluated on an OOD dataset created by merging CheXpert²², NIH²³, SIIM²⁴, PadChest²⁵ and VinDr²⁶. We included all images (both frontal and lateral) and split each dataset into 70% train, 15% validation and 15% test sets. Note that only MIMIC-CXR and CheXpert have patient race information available, and we extracted race (and other attributes) following established protocols⁶⁹. For MIMIC-CXR, demographic information was obtained by merging with MIMIC-IV⁷⁰. For CheXpert, separate race labels were obtained from the Stanford Center for Artificial Intelligence in Medicine & Imaging (<https://aimi.stanford.edu/>) website. Where applicable, we dropped patients with missing values for any attribute.

For all datasets, we excluded samples where the corresponding patient has missing age or sex. For ODIR and ISIC, we dropped samples from patients younger than 18 years and older than 80 years due to small sample sizes (that is, smaller than 3% of the total dataset).

Owing to computational constraints, we mainly chose four prediction tasks for CXRs (that is, ‘No Finding’, ‘Effusion’, ‘Cardiomegaly’ and ‘Pneumothorax’). We selected these tasks for several reasons: (1) diversity in presentation: ‘Effusion’, ‘Cardiomegaly’ and ‘Pneumothorax’ each present distinctively and occur in different locations on a CXR, allowing for a comprehensive evaluation across varied pathologies and underlying causes; (2) prevalence in clinical and research settings: these labels are not only common in clinical practice but also frequently studied in prior academic work^{71,63} and used in commercial diagnostic systems⁷¹; and (3) performance and fairness considerations: these labels are among those with both the highest diagnostic accuracy and substantial fairness gaps on MIMIC-CXR, making them particularly relevant for exploring the relationship between model performance and fairness^{71,2}.

We scaled all images to 224×224 for input to the model. We applied the following image augmentations during training only: random flipping of the images along the horizontal axis, random rotation of up to 10° and a crop of a random size (70–100%) and a random aspect ratio (3/4 to 4/3).

Evaluation methods

To evaluate the performance of disease classification in medical imaging, we used the following metrics: AUROC, TPR, TNR and ECE.

The TPR and TNR are calculated as (FN, false negative; FP, false positive; TP, true positive; TN, true negative):

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

When reporting the sensitivity and specificity, we followed previous work^{12,72} in selecting the threshold that maximizes the F1 score. This threshold optimization procedure is conducted separately for each dataset, task, algorithm and attribute combination. We followed standard procedures to calculate the 95% CI for sensitivity and specificity.

We also reported AUC, which is the area under the corresponding ROC curves showing an aggregate measure of detection performance. Finally, we report the expected calibration error (ECE)⁷³, which we computed using the netcal library⁷⁴.

Assessing the fairness of machine learning models. To assess the fairness of machine learning models, we evaluated the metrics described above for each demographic group as well as the difference in the value of the metric between groups. Equality of TPR and TNR between demographic groups is known in the algorithmic fairness

literature as equal odds⁷⁵. As the models that we studied in this work are likely to be used as screening or triage tools, the cost of an FP may be different from the cost of an FN. In particular, for ‘No Finding’ prediction, FPs (corresponding to underdiagnosis¹²) would be more costly than FNs, and so we focused on the FPR (or TNR) for this task. For all remaining disease prediction tasks, we focused on the FNR (or TPR) for the same reason. Equality in one of the class-conditioned error rates is an instance of equal opportunity³⁰.

Finally, we also examined the per-group ECE and ECE gap between groups. Note that zero ECE for both groups (that is, calibration per group) implies the fairness definition known as sufficiency of the risk score⁷⁵. We emphasize that differences in calibration between groups is a significant source of disparity, as consistent under-estimation or over-estimation of risk for a particular group could lead to under-treatment or over-treatment for that group at a fixed operating threshold relative to the true risk⁷⁶.

Quantifying the distribution shifts

We examined and quantified the types and degrees of distribution shifts in both ID and OOD settings in this study. Inspired by previous work^{46,77}, we performed a series of hypothesis tests to determine if there were significant statistical differences in distributions between demographic groups and across different pairs of datasets. All *P* values were adjusted for multiple testing using Bonferroni correction⁷⁸.

ID setting. We studied the following distribution shifts in the ID setting:

Prevalence shift: $P(Y|A)$. For binary outcomes *Y* across groups, we calculated the total variational distance between the probability distributions of *Y* conditioned on different groups and used a two-sample binomial proportion test, where the null hypothesis corresponds to $P(Y|A = a_1) = P(Y|A = a_2)$:

$$d_Y(a_1, a_2) = d_{TV}(P(Y|A = a_1), P(Y|A = a_2))$$

Representation shift: $P(X|A)$. When comparing the distribution of the input images *X*, we first encoded them into representations derived from a frozen foundation model *f* that is trained in a self-supervised manner on diverse CXR datasets^{79,80}. We then used the mean maximum discrepancy (MMD) distance and a permutation-based hypothesis test following ref. 81 to test if demographic groups differed statistically in their distribution of representations:

$$d_X(a_1, a_2) = d_{MMD}(P(f(X)|A = a_1), P(f(X)|A = a_2))$$

OOD setting. We studied the following distribution shifts in the OOD setting (the null hypothesis is $P_{ID}(\cdot) = P_{OOD}(\cdot)$):

Label shift: $P(Y)$. We calculated the total variational distance between the probability distributions of binary outcomes *Y* across ID and OOD datasets using a two-sample binomial proportion test:

$$d_Y = d_{TV}(P_{ID}(Y), P_{OOD}(Y))$$

Prevalence shift: $P(Y|A = a)$. We similarly evaluated the distance between the distributions of *Y* conditioned on specific demographic subgroups (*A*) between ID and OOD datasets:

$$d_{Y|A}(a) = d_{TV}(P_{ID}(Y|A = a), P_{OOD}(Y|A = a))$$

Covariate shift: $P(X)$. We again encoded *X* into representations derived from a frozen foundation model *f* and then used the MMD distance and a permutation-based hypothesis test⁸¹ to examine if ID and OOD datasets differed statistically in their distribution of representations:

$$d_X = d_{MMD}(P_{ID}(f(X)), P_{OOD}(f(X)))$$

Representation shift: $P(X|A = a)$. Similarly, we calculated the MMD distance conditioned on subgroup A to evaluate shifts in the representation space:

$$d_{X|A}(a) = d_{\text{MMD}}(P_{\text{ID}}(f(X)|A = a), P_{\text{OOD}}(f(X)|A = a))$$

We provide additional results on quantifying various distribution shifts in both ID and OOD settings in the Supplementary Information (Supplementary Tables 1–3).

Training details

We trained DenseNet-121 (ref. 33) models on each task, initializing with ImageNet⁸² pre-trained weights. We evaluated six algorithms: ERM³⁴, ReSample³⁵, GroupDRO³⁶, DANN³⁷, CDANN³⁸ and MA³⁹.

For each combination of task, algorithm and demographic attribute, we conducted a random hyperparameter search⁸³ with 12 runs. During training, for a particular attribute, we evaluated the validation set worst-group validation AUROC every 1,000 steps and early stopped if this metric had not improved for five evaluations. We tuned the learning rate and weight decay for all algorithms and also tuned algorithm-specific hyperparameters as mentioned in the original works. We selected the hyperparameter setting that maximized the worst-attribute validation AUROC. CIs were computed as the standard deviation across three different random seeds for each hyperparameter setting.

We also explored a multi-label training setup, where models were trained simultaneously on 14 binary labels available in MIMIC-CXR⁷. We followed the same experimental protocol as outlined in the main paper, including hyperparameter tuning and model selection. Our findings in the multi-label setup mirrored those seen in the binary task setup (Supplementary Figs. 2 and 3).

To obtain the level of demographic encoding within representations (Fig. 2), we first computed representations using a trained disease prediction model. We froze these representations and trained a multi-class multi-nomial logistic regression model to predict the demographic group using the training set using the scikit-learn library⁸⁴. We varied the L2 regularization strength between 10^{-3} and 10 and selected the model with the best macro-averaged AUROC on the validation set. We report the macro-averaged AUROC on the test set.

Decomposing OOD fairness

Here we present a first approach toward decomposing the fairness gap in an OOD environment as a function of the ID fairness gap and the impact that the distribution shift has in each group. In particular, let D_{src} and D_{tar} be the source and target datasets, respectively. Let $g \in G$ be a particular group from a set of groups. Let $L_f(g, D)$ be an evaluation metric for a model f , which is decomposable over individual samples—that is, $L_f(g, D) = L_f(g, D) = \sum_{(x,y,g) \in D: g=g} l(f(x), y)$. Examples of such metrics are the accuracy, TPR or TNR. Then, we can decompose:

$$L_f(g_1, D_{\text{tar}}) - L_f(g_2, D_{\text{tar}}) = [L_f(g_1, D_{\text{src}}) - L_f(g_2, D_{\text{src}})] \\ + [L_f(g_2, D_{\text{src}}) - L_f(g_2, D_{\text{tar}})] - [L_f(g_1, D_{\text{src}}) - L_f(g_1, D_{\text{tar}})].$$

The left-hand term is the fairness gap in the OOD environment, and the three terms on the right are (1) the fairness gap in the ID data, (2) the impact of the distribution shift on g_2 and (3) the impact of the distribution shift on g_1 . We note that, to achieve a low fairness gap in the OOD environment, it is important not only to minimize the ID fairness gap (term 1) but also to minimize the difference in how the distribution shift impacts each group (term 2 – term 3).

Evaluation with different medical imaging modalities

In addition to radiology, we also examined medical AI applications in dermatology and ophthalmology to corroborate our findings. Specifically, Extended Data Fig. 1 shows the results for dermatological imaging. We used the ISIC dataset²⁸, which contains 32,259 images sourced

from multiple international sites. We focused on the ‘No Finding’ task, taking into account ‘sex’ and ‘age’ as the sensitive demographic attributes (Extended Data Fig. 1a). Similar to our observations in radiology, we identified fairness gaps within subgroups based on age and sex (Extended Data Fig. 1b), although these disparities were less significant than those observed in CXR assessments (for example, fairness gaps smaller than 2%). This was further confirmed by the Pareto front plot, where most models, including ERM, could achieve a good performance–fairness tradeoff (Extended Data Fig. 1c).

We extended our analysis to ophthalmology images, specifically focusing on retinopathy detection, using the ODIR dataset²⁹ with 6,800 images (Extended Data Fig. 2). The task that we considered was ‘Retinopathy’, with ‘sex’ and ‘age’ being used as demographic attributes (Extended Data Fig. 2a). Notably, significant subgroup fairness gaps were observed in age (43% FNR gap between groups ‘60–80’ and ‘18–40’). In contrast, the fairness gap based on sex was less significant, with a 3% FNR difference between ‘female’ and ‘male’ subgroups. We further verified the Pareto front for both attributes, where similar observations hold that algorithms for fixing demographic shortcuts could improve ID fairness while incurring minimal detriments to performance (measured in AUROC).

Analysis on underdiagnosis versus overdiagnosis

In evaluating fairness metrics, our primary study centered on underdiagnosis, specifically the disparities in FPR for ‘No Finding’ and discrepancies in FNR for other conditions. However, an alternative approach involves focusing on overdiagnosis, defined as variances in FNR for ‘No Finding’ and differences in FPR for other diseases. We present findings between their relationship in Extended Data Fig. 3. An analysis spanning two datasets (MIMIC and CheXpert) and various tasks revealed a consistent pattern: larger gaps in underdiagnosis tend to correspond with more significant overdiagnosis discrepancies. Nonetheless, certain task and attribute combinations exhibited more complex trends, indicating a necessity for deeper exploration and informed decision-making regarding the most appropriate fairness metrics for critical disease evaluations in practical medical settings.

Direct prediction of demographic attributes

We provide analyses for demographic encoding of attribute information. In the main paper, we analyzed the predictiveness of attributes (for example, age, sex and race) based on the embeddings from a disease classification model. The distinct predictiveness between attributes in these domains could be attributed to the intrinsic characteristics of the datasets or the nature of the conditions being studied. To delve deeper, we conducted an experiment training an ERM model to predict these attributes directly using the dermatology dataset (ISIC), and we show the results in Supplementary Table 4. We observed that certain attributes are indeed less predictive compared to others (that is, age versus sex), suggesting that age may be inherently more challenging to encode within the studied dermatology dataset. Furthermore, the results reveal variations in the predictiveness of age across different subgroups (for example, age groups ‘18–40’ and ‘60–80’ exhibit higher AUROC than the ‘40–60’ group).

Analysis using multi-label models

Prior works⁷ studied CXR classification in the multi-label setting, where a model contains an encoder, followed by an individual linear classification head for each of the downstream tasks. We followed the setup⁷ to study the following 14 binary labels in MIMIC-CXR: Atelectasis, Cardiomegaly, Consolidation, Edema, Enlarged Cardiomediastinum, Fracture, Lung Lesion, Airspace Opacity, No Finding, Effusion, Pleural Other, Pneumonia, Pneumothorax and Support Devices.

We adapted the following methods to the multi-label setting, (1) ERM, (2) DANN and (3) CDANN, and we followed the same experimental protocol in the main paper in terms of hyperparameter tuning.

Note that GroupDRO and ReSample are challenging to adapt to the multi-label setting, as the number of groups is exponential in the number of tasks. For each combination of algorithm and attribute, we selected the multi-label model that maximizes the worst-attribute AUROC, averaged across the 14 tasks.

First, we examined the level of demographic encoding present in the embeddings of the best multi-task ERM model and found that it also encodes a variety of demographic information, similar to the single-label case (Supplementary Fig. 2a). We further showed the fairness gaps of this best multi-label ERM model and observed that a variety of fairness gaps exist and are statistically significant across all tasks (Supplementary Fig. 2b). In addition, we plotted the correlation between the fairness gap and the attribute prediction AUROC, across all trained multi-label models. We found a strong and statistically significant positive correlation among all combinations of task and attribute, similar to the single-label case (Supplementary Fig. 2c).

We also present Pareto plots showing the tradeoff between the fairness gap and overall AUROC across all models, for each combination of task and attribute (Supplementary Fig. 3). Overall, we found that the Pareto fronts for the multi-label models demonstrate similar behavior as the single-task models—where the multi-label ERM exhibits the best overall AUROC but has high fairness gap. In addition, with debiasing methods such as multi-label DANN and CDANN, we are able to achieve fair models with minimal loss in overall AUROC.

Test set rebalancing for prevalence shift

We investigated whether eliminating the prevalence shift in the test set would address the observed fairness gaps. Following prior work⁴⁰, we balanced the test set for multiple attributes—age and race—ensuring that demographic proportions (for example, ‘White’ aged ‘20–40’ versus ‘Black’ aged ‘60–80’) and disease prevalence are uniform across all attribute combinations. This approach aims to eliminate prevalence shifts within the test set. Our findings in Supplementary Fig. 4 suggest that, although test set rebalancing can reduce fairness gaps for certain task and attribute combinations (for example, ‘No Finding’ for ‘race’ and ‘Cardiomegaly’ for ‘age’, as compared to Fig. 2b), there exists significant gaps even after rebalancing the test set, indicating that fairness gaps are influenced by multiple shifts beyond just prevalence shifts.

Statistical analysis

Correlation. To calculate the correlations between variables, we used Pearson correlation coefficients and their associated *P* value (two-sided *t*-test, $\alpha = 0.05$). 95% CI for the Pearson correlation coefficient was calculated.

Increase in OOD fairness gap. One-tailed Wilcoxon rank-sum test ($\alpha = 0.05$) was used to assess the increase in OOD fairness gap compared to oracle models.

CIs. We used non-parametric bootstrap sampling to generate CIs: random samples of size *n* (equal to the size of the original dataset) were repeatedly sampled 1,000 times from the original dataset with replacement. We then estimated the increase in OOD fairness gap compared to oracle using each bootstrap sample ($\alpha = 0.05$).

All statistical analysis was performed with Python version 3.9 (Python Software Foundation).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets used in this study are publicly available. The MIMIC-CXR (<https://www.physionet.org/content/mimic-cxr-jpg/2.1.0/>) and VinDr-CXR (<https://physionet.org/content/vindr-cxr/1.0.0/>) datasets

are available from PhysioNet after completion of a data use agreement and a credentialing procedure. The CheXpert dataset (<https://stanfordmlgroup.github.io/competitions/chexpert/>), along with associated race labels, is available from the Stanford Center for Artificial Intelligence in Medicine & Imaging website (<https://aimi.stanford.edu/>). The ChestX-ray14 (National Institutes of Health) dataset (<https://nihcc.app.box.com/v/ChestXray-NIHCC>) is available to download from the National Institutes of Health Clinical Center. The PadChest dataset (<https://academictorrents.com/details/96eb4b4f92b85929eadfb16761f310a6d04105797>) can be downloaded from the Medical Imaging Databank of the Valencia Region. The SIIM-ACR Pneumothorax Segmentation dataset (<https://www.kaggle.com/c/siim-acr-pneumothorax-segmentation>) can be downloaded from its Kaggle contest page. The ISIC 2020 dataset (<https://challenge.isic-archive.com/data/#2020>) can be downloaded from the SIIM-ISIC Melanoma Classification Challenge page. The ODIR dataset (<https://odir2019.grand-challenge.org/dataset/>) can be obtained from the ODIR 2019 challenge hosted by Grand Challenges.

Code availability

Code that supports the findings of this study is publicly available with an open-source license at <https://github.com/YyzHarry/shortcut-ood-fairness>.

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Author contributions

Y.Y., H.Z. and M.G. conceived and designed the study. Y.Y. and H.Z. performed data collection, processing and experimental analysis. Y.Y., H.Z., J.W.G., D.K. and M.G. interpreted experimental results and provided feedback on the study. Y.Y., H.Z., J.W.G. and M.G. wrote the original manuscript. M.G. supervised the research. All authors reviewed and approved the manuscript.

Competing interests

D.K. is a co-founder of Emerald Innovations, Inc. and serves on the scientific advisory board of Janssen and on the data and analytics advisory board of Amgen. The other authors declare no competing interests.

Additional information

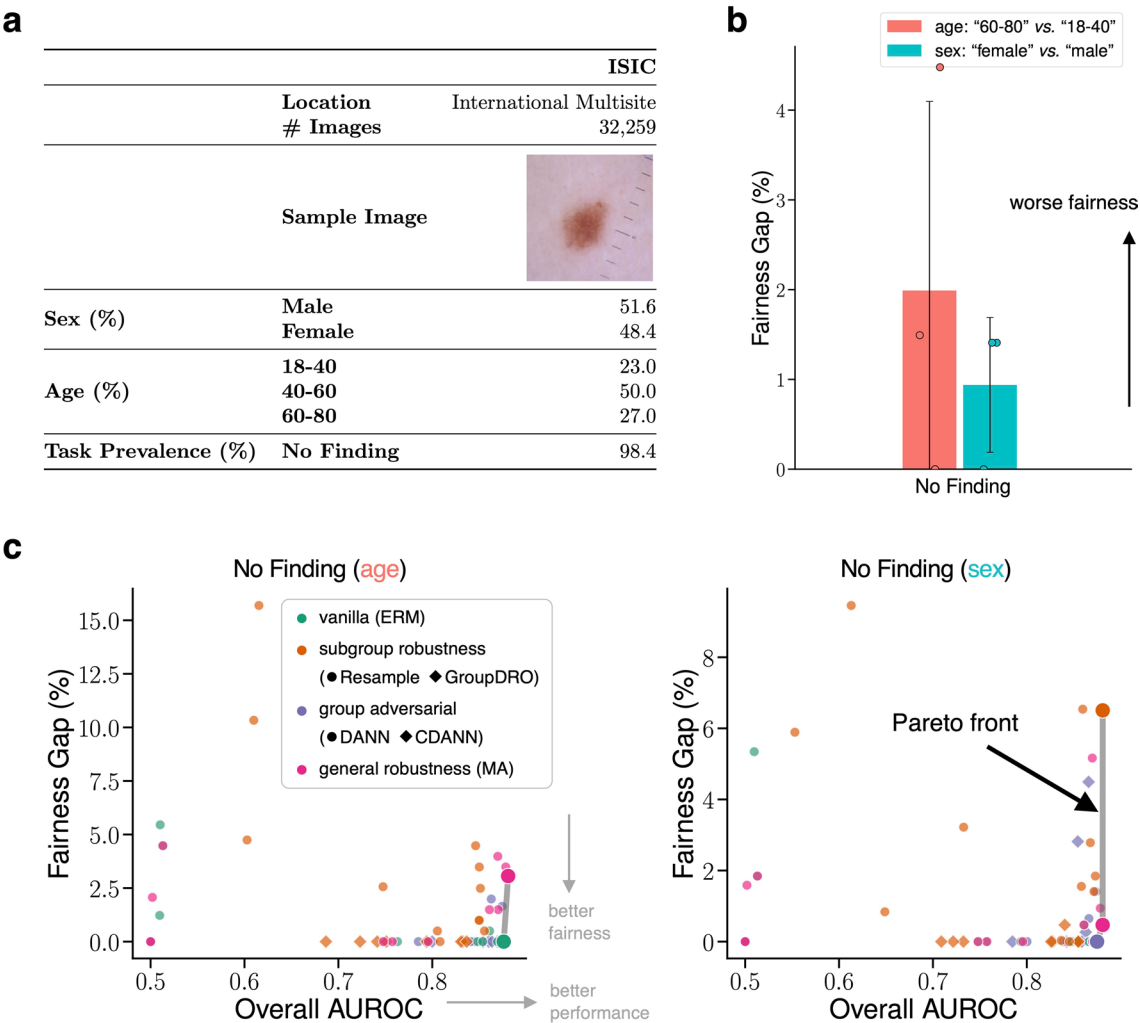
Extended data is available for this paper at <https://doi.org/10.1038/s41591-024-03113-4>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41591-024-03113-4>.

Correspondence and requests for materials should be addressed to Yuzhe Yang.

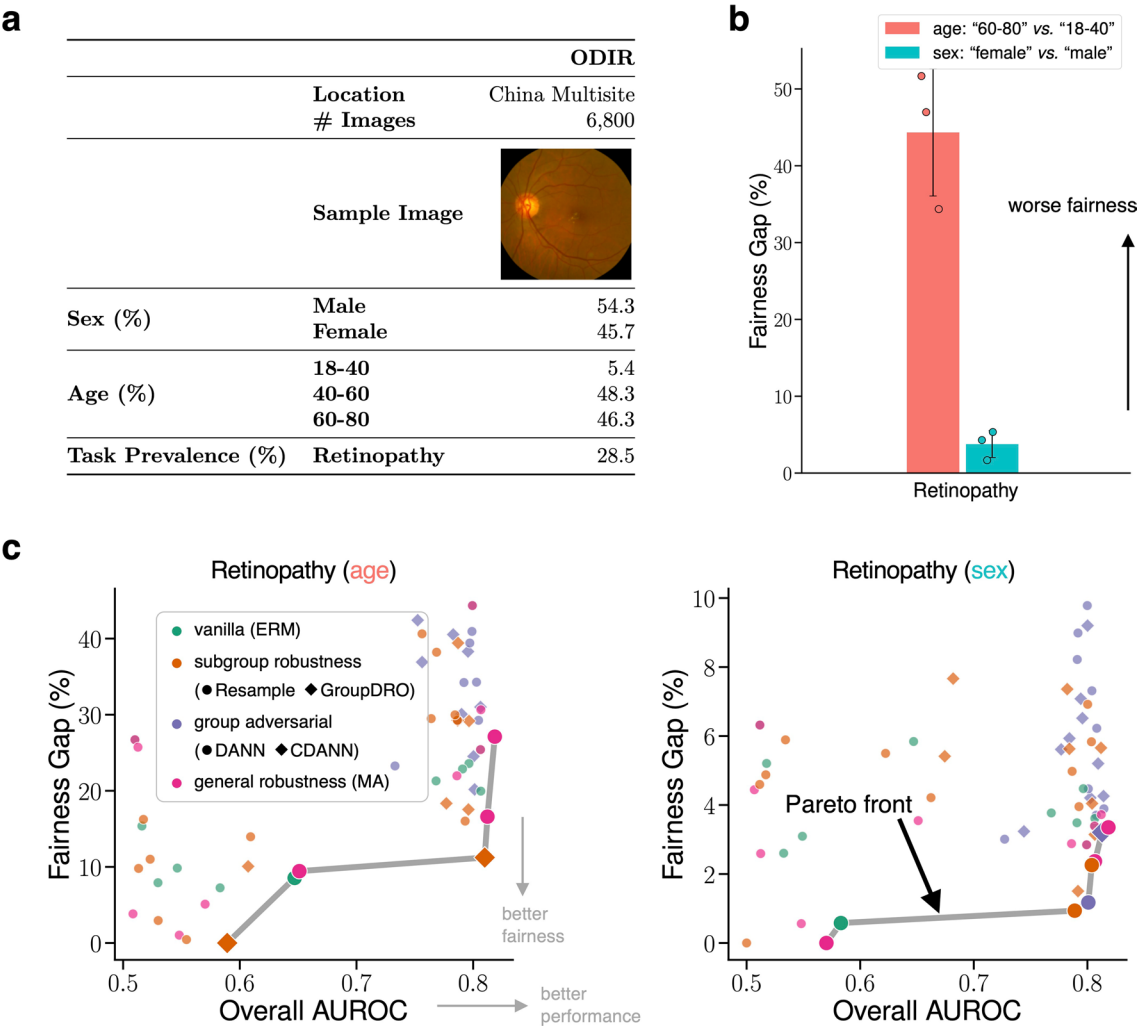
Peer review information *Nature Medicine* thanks Melanie Roschewitz and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Lorenzo Righetto, in collaboration with the *Nature Medicine* team.

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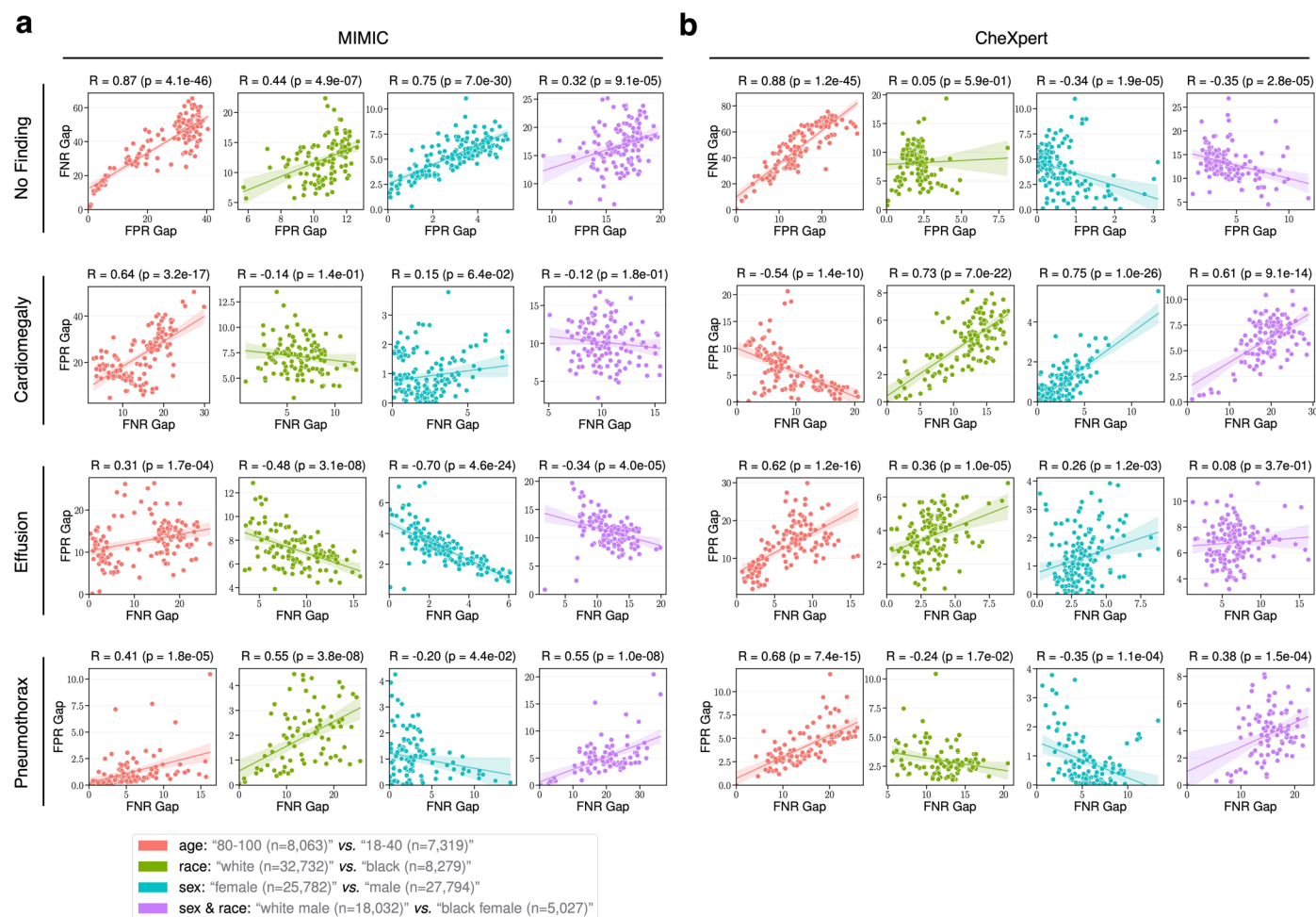
Extended Data Fig. 1 | Evaluation results for models on the ISIC dataset for No Finding prediction. **a**, Dataset statistics. **b**, Subgroup fairness gaps of the ERM model as defined by FPR. Each subgroup contains at least 100 samples for analysis (age: subgroup ‘60–80’ vs ‘18–40’; sex: subgroup ‘female’ vs ‘male’). We find that disparities in FPR are small and statistically insignificant in the case of

age. Each bar and its error bar indicate the mean and standard deviation across 3 independent runs. **c**, Trade-off between the fairness gap and overall AUROC for all trained models, evaluated against sensitive attribute age and sex, respectively. We find that most models, including ERM, achieve a good fairness-performance tradeoff.



Extended Data Fig. 2 | Evaluation results for models on the ODIR dataset for Retinopathy prediction. **a**, Dataset statistics. **b**, Subgroup fairness gaps of the ERM model as defined by FNR. Each subgroup contains at least 100 samples for analysis (age: subgroup ‘60–80’ vs ‘18–40’; sex: subgroup ‘female’ vs ‘male’). We find a significant FNR gap between age groups. Each bar and its error bar

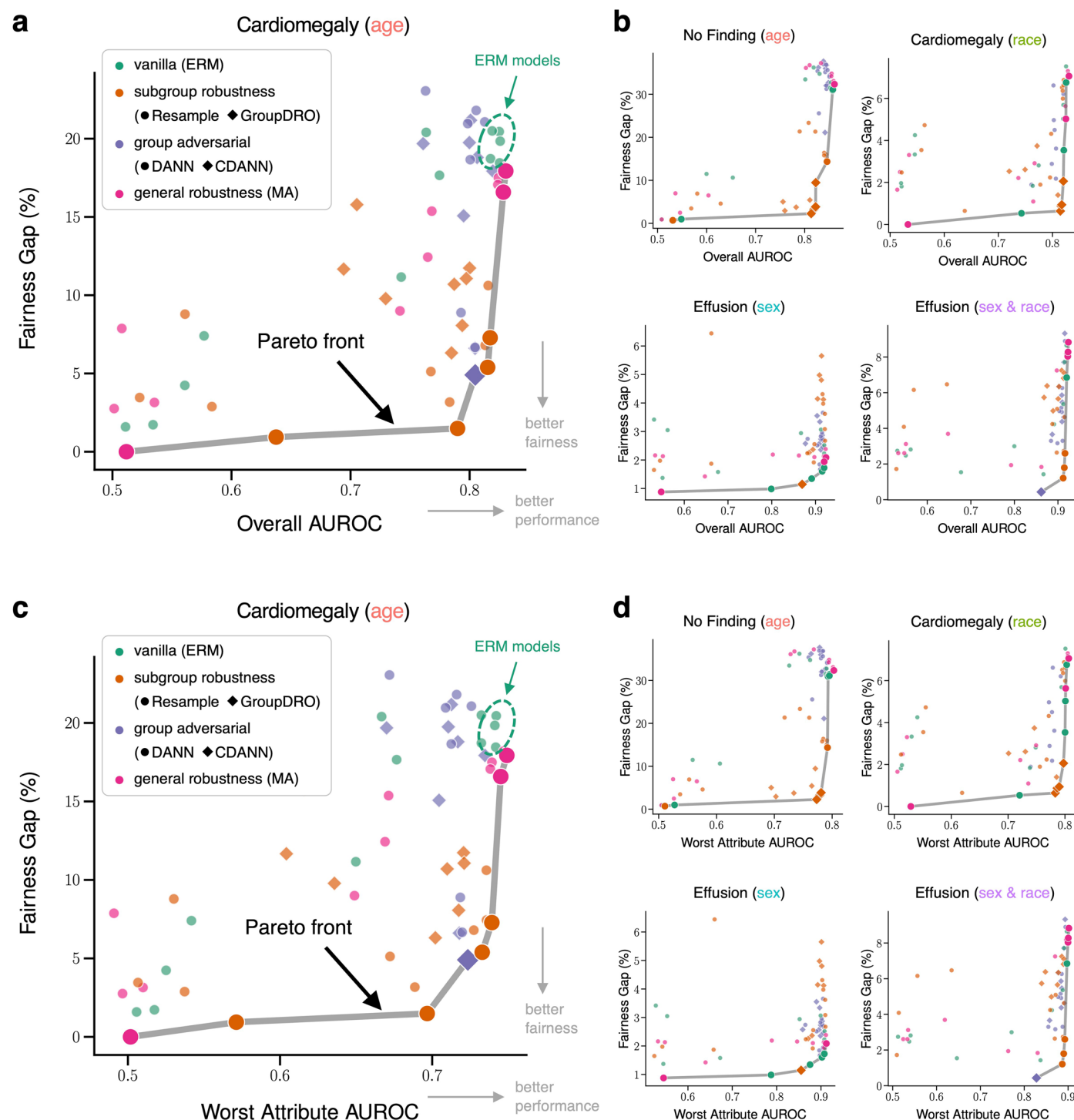
indicate the mean and standard deviation across 3 independent runs. **c**, Trade-off between the fairness gap and overall AUROC for all trained models, evaluated against sensitive attribute age and sex, respectively. We find, similar to the chest X-ray setting, that algorithms for fixing demographic shortcuts could improve in-distribution fairness while incurring minimal detriments to performance.



Extended Data Fig. 3 | Trade-offs between the FPR gap and the FNR gap for each task and attribute, for models trained on MIMIC-CXR or CheXpert and evaluated on the same dataset for (a) No Finding and (b) Effusion prediction.

We evaluate these metrics across age ('80-100' vs '18-40'), sex ('female' vs 'male'), race ('White' vs 'Black'), and the intersection of sex and race ('White

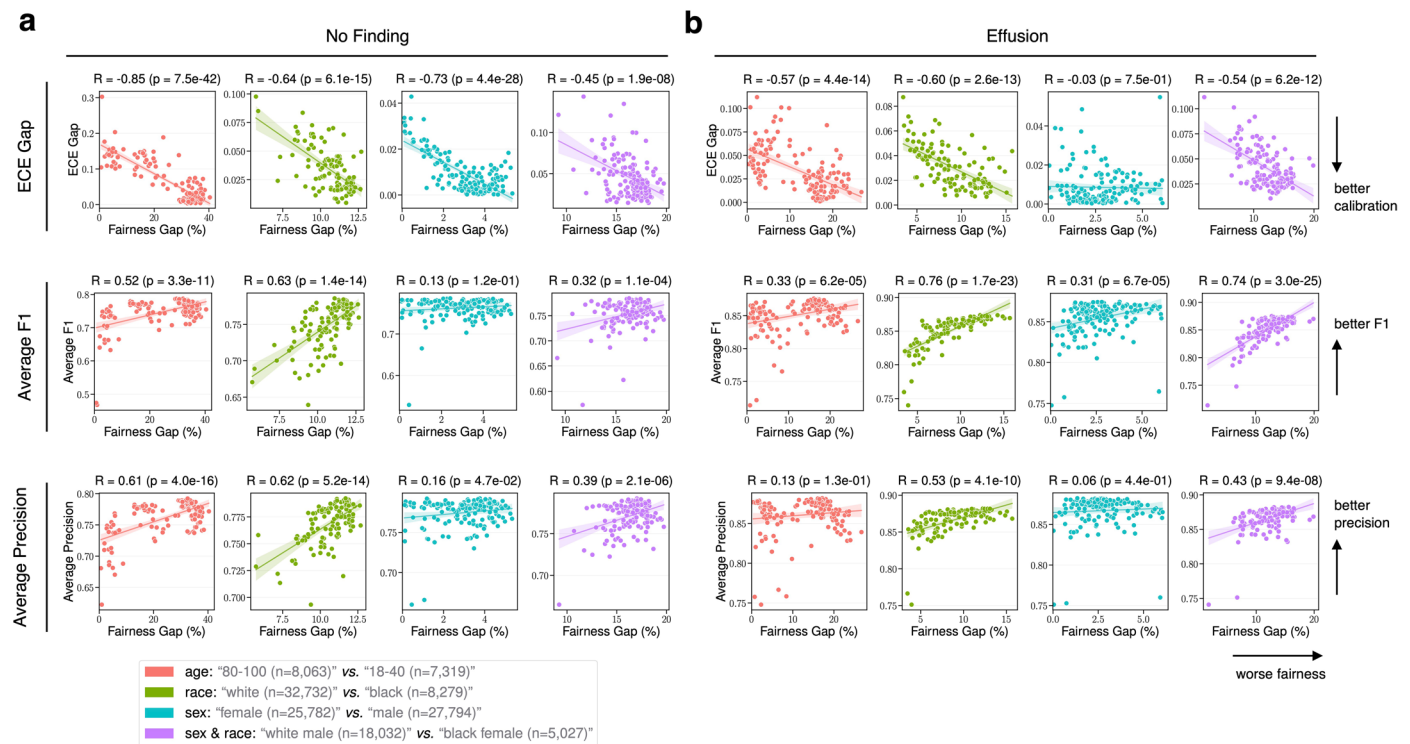
male' vs. 'Black female'). We find for the most part, there is a positive correlation, indicating that fairer models achieve fairness with respect to both FPR and FNR (that is, equal odds). All p values are calculated using two-sided t-test. The center line and the shadow denote the mean and 95% CI, respectively.



Extended Data Fig. 4 | Algorithms for removing demographic shortcuts mitigate in-distribution fairness gaps and maintain performance.

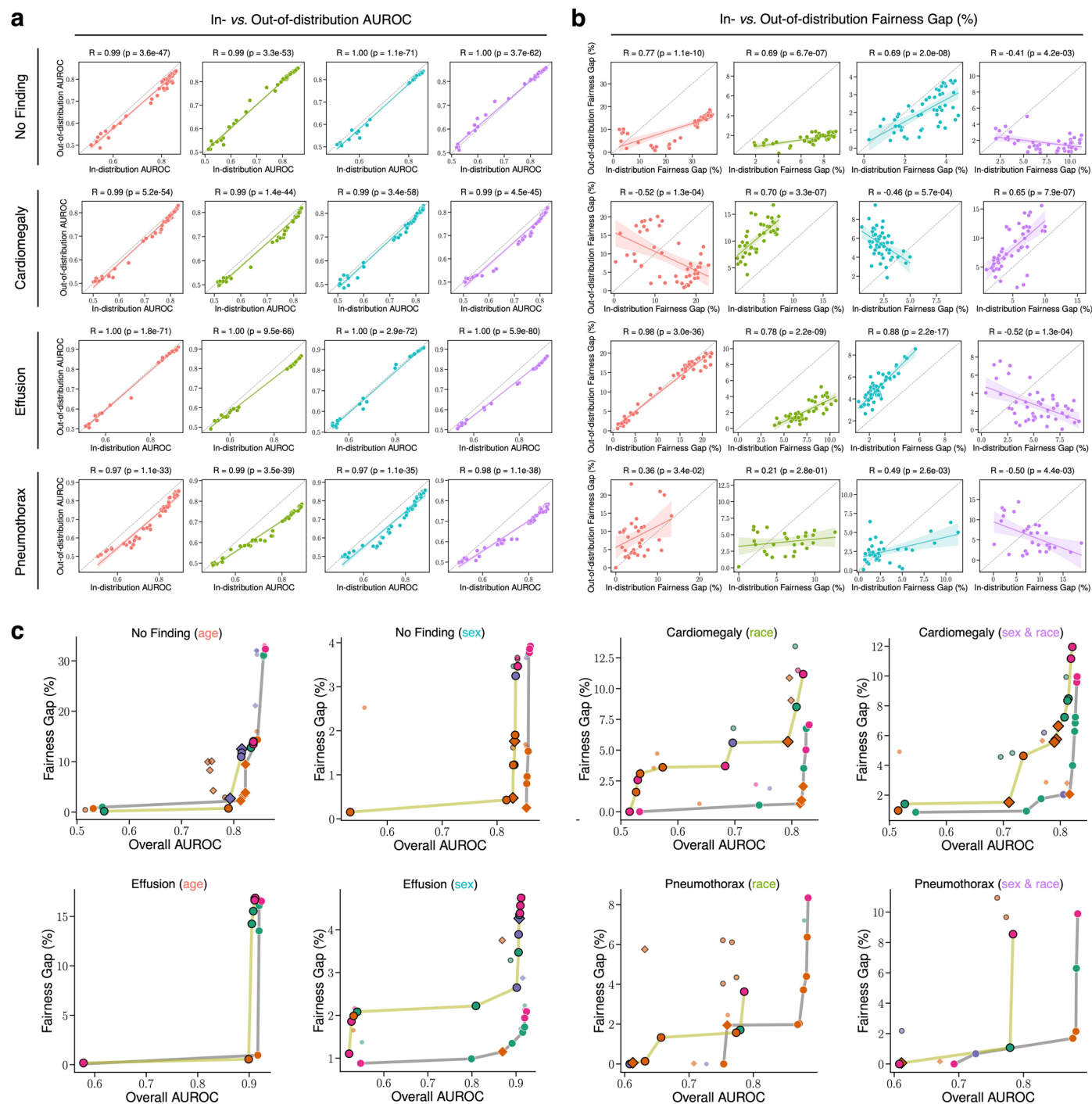
a, b, Trade-off between the fairness gap and overall AUROC for all trained models. **c, d**, Trade-off between the fairness gap and the worst-group AUROC

for all trained models. Each plot represents a specific disease prediction task (for example, Cardiomegaly) with a specific attribute (for example, age). In each case, we plot the Pareto front, the best achievable fairness gap with a minimum constraint on the performance.



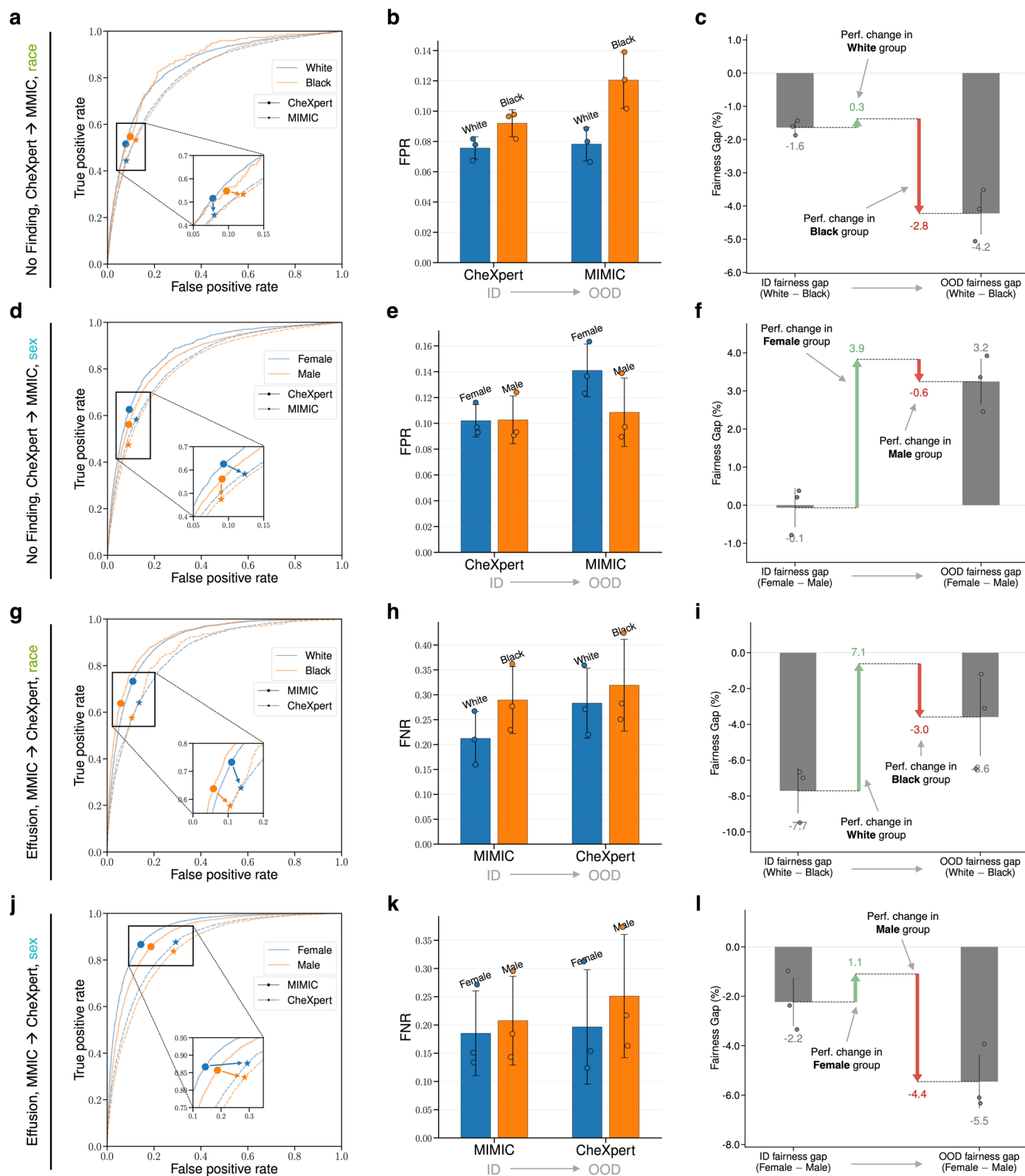
Extended Data Fig. 5 | Trade-offs between the fairness gap and the expected calibration error (ECE) gap, average F1, and average precision, for models trained and evaluated on MIMIC-CXR for (a) No Finding and (b) Effusion prediction. We evaluate these metrics across age ('80-100' vs '18-40'), sex ('female' vs 'male'), race ('White' vs. 'Black'), and the intersection of sex and

race ('White male' vs. 'Black female'). We find that fairer models tend to exhibit larger ECE gaps, worse average F1, and worse average precision, indicating the undesirable tradeoff between fairness with other performance and calibration metrics. All p values are calculated using two-sided t-test. The center line and the shadow denote the mean and 95% CI, respectively.



Extended Data Fig. 6 | The transfer of performance (overall AUROC) and fairness between the ID (MIMIC-CXR) and OOD datasets. a, The OOD performance versus the ID performance for each task and attribute combination. **b,** The OOD fairness versus the ID fairness for each task and attribute

combination. **c,** The Pareto front between fairness and performance for ID and OOD, finding that models that are Pareto optimal ID often do not maintain Pareto optimality OOD. All p values are calculated using two-sided t-test. The center line and the shadow denote the mean and 95% CI, respectively.



Extended Data Fig. 7 | Examining the behavior of specific ERM models transferred from ID setting to OOD setting. a, d, g, j. The receiver operating characteristic (ROC) curves for each group and each dataset, marking the operating point of the model. **b, e, h, k.** This shift in the operating point results in a change in the FPR and FNR values of subgroups on the OOD dataset. Each bar

and its error bar indicate the mean and standard deviation across 3 independent runs. **c, f, i, l.** We decompose the OOD fairness gap as a function of the ID fairness gap, and the change in FPR for each of the groups. Each bar and its error bar indicate the mean and standard deviation across 3 independent runs.

Extended Data Table 1 | Prevalence rates for each demographic subgroup of the datasets used for training models in this study

		MIMIC				CheXpert			
		Cardiomegaly	No Finding	Effusion	Pneumothorax	Cardiomegaly	No Finding	Effusion	Pneumothorax
Sex (%)	Female	15.1	42.6	18.9	2.8	11.6	10.2	38.8	8.3
	Male	14.8	37.2	21.1	4.0	12.4	9.9	38.4	9.0
Race (%)	Asian	16.6	36.0	24.2	5.4	12.7	10.4	40.5	9.8
	Black	17.6	44.3	13.4	1.8	19.6	11.7	31.7	5.8
	White	15.5	34.6	24.0	4.0	11.5	9.4	39.4	9.1
	Other	11.1	52.5	12.6	2.5	11.7	10.8	37.6	8.0
Age (%)	18-40	6.8	64.0	8.1	3.6	9.1	20.5	27.0	12.5
	40-60	11.4	46.5	15.0	3.0	10.1	12.4	36.2	8.6
	60-80	17.6	32.5	23.9	3.8	12.4	7.0	42.3	8.9
	80-100	22.9	23.3	31.0	3.0	17.9	3.7	44.2	5.0
Intersection (%)	Asian Female	16.9	38.6	22.7	4.7	12.5	10.7	40.5	9.8
	Asian Male	16.4	33.6	25.5	6.1	12.8	10.1	40.4	9.8
	Black Female	18.3	46.6	13.0	1.5	20.9	11.5	31.8	4.7
	Black Male	16.7	41.0	13.9	2.2	18.2	11.9	31.6	6.8
	White Female	15.5	36.3	23.5	3.5	10.2	9.4	39.9	9.0
	White Male	15.4	33.3	24.4	4.4	12.4	9.4	39.0	9.2
	Others Female	10.7	57.2	10.9	1.7	12.1	11.4	37.6	7.0
	Others Male	11.5	48.1	14.2	3.3	11.4	10.4	37.6	8.7

Extended Data Table 2 | Distribution shift quantification in the ID settings between demographic groups on MIMIC-CXR

a Shifts in Y : $P(Y|A = a_1)$ vs. $P(Y|A = a_2)$

Attribute	Group 1	Group 2	Distance	p value [†]
Sex	Female	Male	0.054	***
Age	40-60	18-40	0.187	***
	80-100	40-60	0.221	***
	80-100	60-80	0.083	***
	80-100	18-40	0.408	***
	60-80	40-60	0.138	***
	60-80	18-40	0.325	***
Race	White	Black	0.097	***
	White	Other	0.176	***
	White	Asian	0.005	not significant
	Black	Other	0.079	***
	Black	Asian	0.092	***
	Asian	Other	0.171	***

† Bonferroni correction for multiple testing. *** indicates $p < 0.001$.

b Shifts in X : $P(X|A = a_1)$ vs. $P(X|A = a_2)$

Attribute	Group 1	Group 2	Distance	p value [†]
Sex	Female	Male	0.005	***
Age	40-60	18-40	0.048	***
	80-100	40-60	0.076	***
	80-100	60-80	0.013	***
	80-100	18-40	0.221	***
	60-80	40-60	0.022	***
	60-80	18-40	0.133	***
Race	White	Black	0.015	***
	White	Other	0.036	***
	White	Asian	0.002	***
	Black	Other	0.008	***
	Black	Asian	0.012	***
	Asian	Other	0.025	***

† Bonferroni correction for multiple testing. *** indicates $p < 0.001$.

a, Prevalence shift $P(Y|A)$ was derived using the total variational distance between the probability distributions of Y conditioned on different groups. P values were computed using a two-sided proportion z-test. **b**, Representation shift $P(X|A)$ was derived by first encoding input into representations from a frozen foundation model f (that is, MedCLIP⁷⁶) and then computing the MMD distance with a Gaussian kernel⁸¹. P values were computed using a two-sided permutation test using this distance as the test statistic⁸¹. All P values were adjusted for multiple testing using Bonferroni correction⁷⁸.

Extended Data Table 3 | Distribution shift quantification in the OOD settings between different datasets

a Shifts in Y : $P_{ID}(Y)$ vs. $P_{OOD}(Y)$				b Shifts in X : $P_{ID}(X)$ vs. $P_{OOD}(X)$			
ID dataset	OOD dataset	Distance	p value [†]	ID dataset	OOD dataset	Distance	p value [†]
MIMIC	CheXpert	0.299	***	MIMIC	CheXpert	0.044	***
	NIH	0.142	***		NIH	0.020	***
	PadChest	0.045	***		PadChest	0.225	***
	VinDr	0.014	not significant		VinDr	0.042	***
	Combined	0.109	***		SIIM	0.027	***
CheXpert	MIMIC	0.299	***	CheXpert	MIMIC	0.044	***
	NIH	0.442	***		NIH	0.076	***
	PadChest	0.254	***		PadChest	0.267	***
	VinDr	0.313	***		VinDr	0.122	***
	Combined	0.191	***		SIIM	0.049	***
† Bonferroni correction for multiple testing. *** indicates $p < 0.001$.				† Bonferroni correction for multiple testing. *** indicates $p < 0.001$.			
c Shifts in $Y A$: $P_{ID}(Y A=a)$ vs. $P_{OOD}(Y A=a)$ [‡]				d Shifts in $X A$: $P_{ID}(X A=a)$ vs. $P_{OOD}(X A=a)$ [‡]			
Attribute	Group	Distance	p value	Attribute	Group	Distance	p value
Sex	Female	0.326	***	Sex	Female	0.050	***
	Male	0.274	***		Male	0.035	***
Age	80-100	0.201	***	Age	80-100	0.023	***
	60-80	0.255	***		60-80	0.035	***
	40-60	0.338	***		40-60	0.063	***
	18-40	0.443	***		18-40	0.102	***
Race	White	0.256	***	Race	White	0.035	***
	Black	0.324	***		Black	0.048	***
	Asian	0.252	***		Asian	0.037	***
	Other	0.414	***		Other	0.094	***
‡ ID and OOD datasets here are MIMIC and CheXpert, respectively.				‡ ID and OOD datasets here are MIMIC and CheXpert, respectively.			

a, Label shift $P(Y)$ was derived using the total variational distance between the probability distributions of Y between ID and OOD datasets. P values were computed using a two-sided proportion z-test. **b**, Covariate shift $P(X)$ was derived by first encoding input into representations from a frozen foundation model f (that is, MedCLIP⁷⁹) and then computing the MMD distance with a Gaussian kernel⁸¹ between ID and OOD datasets. P values were computed using a two-sided permutation test using this distance as the test statistic⁸¹. **c**, Prevalence shift $P(Y|A=a)$ was derived using the total variational distance conditioned on specific subgroups between ID and OOD datasets. P values were computed using a two-sided proportion z-test. **d**, Representation shift $P(X|A=a)$ was derived using the MMD distance on specific subgroups between ID and OOD datasets⁸¹. P values were computed using a two-sided permutation test using this distance as the test statistic⁸¹. All P values were adjusted for multiple testing using Bonferroni correction⁷⁸.

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
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☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software was used for data collection.
Data analysis	Code was written using Python version 3.9 (Python Software Foundation), with pytorch v1.12.0 and scikit-learn v1.2.2. Code is available for non-commercial research purpose, and is publicly available with an open-source license at https://github.com/YyzHarry/shortcut-ood-fairness .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All datasets used in this study are publicly available. The MIMIC-CXR and VinDr-CXR datasets are available from PhysioNet after the completion of a data use agreement and a credentialing procedure. The CheXpert dataset, along with associated race labels, is available from the Stanford AIMI website. The ChestX-ray14

(NIH) dataset is available to download from the National Institute of Health Clinical Center. The PadChest dataset can be downloaded from the Medical Imaging Databank of the Valencia Region. The SIIM-ACR Pneumothorax Segmentation dataset can be downloaded from its Kaggle contest page. The ISIC 2020 dataset can be downloaded from the SIIM-ISIC Melanoma Classification Challenge page. The ODIR dataset can be obtained from the ODIR 2019 challenge hosted by Grand Challenges.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

For all datasets, self-reported sex was used for fairness analyses. The datasets used in this study have sex distributions publicly reported (Table 1, Extended Data Figure 1a & 2a). We report the results stratified by sex to assess the model performance on different subgroups, and explore the ability of the trained models to predict sex from medical images in the datasets used.

- MIMIC-CXR: 170,698 images (female) and 186,469 images (male)
- CheXpert: 90,576 images (female) and 132,216 images (male)
- NIH: 48,780 images (female) and 63,340 images (male)
- SIIM: 5,161 images (female) and 6,421 images (male)
- PadChest: 72,858 images (female) and 71,620 images (male)
- VinDr: 2,741 images (female) and 3,613 images (male)
- ISIC: 15,978 images (female) and 17,080 images (male)
- ODIR: 3,196 images (female) and 3,766 images (male)

Reporting on race, ethnicity, or other socially relevant groupings

Self-reported race was used for fairness analyses in MIMIC-CXR and CheXpert. Race information is not available in NIH, SIIM, PadChest, VinDr, ISIC, and ODIR. We report the results stratified by race to assess the model performance on different subgroups, and explore the ability of the trained models to predict race from medical images in the datasets used.

- MIMIC-CXR: 11,121 images (Asian), 55,611 images (Black), 218,037 images (White), and 72,398 images (Others)
- CheXpert: 23,298 images (Asian), 11,970 images (Black), 125,624 images (White), and 61,900 images (Others)

Population characteristics

See above, as well as Table 1 in the paper.

Recruitment

No participants were recruited for this retrospective study.

Ethics oversight

The study used data collected retrospectively. Approval of a study protocol was not needed.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

We utilized seven datasets from the radiology domain, which encompass the set of publicly available chest X-ray datasets that have been widely used in prior work. These datasets span three continents, and allow us to evaluate model performance and fairness under real-world distribution shifts. In addition, we utilized one dataset each from the ophthalmology and dermatology domains, to validate our in-distribution findings on other medical imaging modalities. Sample sizes are the following:

Radiology: In total, we used 854,493 chest X-ray images for evaluation, coming from 5 internationally-sourced datasets: MIMIC (357,167 images), CheXpert (222,792 images), NIH (112,120 images), SIIM (11,582 images), PadChest (144,478 images), and Vin Dr (6,354 images).
Ophthalmology: We used ophthalmology images from ODIR dataset (6,800 images).
Dermatology: We used dermatology images from ISIC dataset (32,259 images).

Data exclusions

For all datasets, we exclude samples where the corresponding patient has missing age or sex. For ODIR and ISIC, we drop samples from patients younger than 18 and older than 80 due to small sample sizes (i.e., smaller than 3% of the total dataset).

No other data were excluded.

Replication

The code and data used to evaluate the findings are publicly available (as detailed in the Data-availability and Code-availability statements), can be used to replicate the findings.

Randomization

Randomization was used to create the train, validation, and test sets for each of the datasets used in the study.

Blinding

As the study contained no human-subject evaluations or interventions, and only profiles computational models on a fixed retrospectively collected dataset, no blinding was necessary.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

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☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

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Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.